Markov Chains

4

4.1 Introduction

Consider a process that has a value in each time period. Let X_n denote its value in time period n, and suppose we want to make a probability model for the sequence of successive values $X_0, X_1, X_2 \ldots$. The simplest model would probably be to assume that the X_n are independent random variables, but often such an assumption is clearly unjustified. For instance, starting at some time suppose that X_n represents the price of one share of some security, such as Google, at the end of n additional trading days. Then it certainly seems unreasonable to suppose that the price at the end of day $n+1$ is independent of the prices on days $n, n-1, n-2$ and so on down to day 0. However, it might be reasonable to suppose that the price at the end of trading day $n+1$ depends on the previous end-of-day prices only through the price at the end of day n. That is, it might be reasonable to assume that the conditional distribution of X_{n+1} given all the past end-of-day prices $X_n, X_{n-1}, \ldots, X_0$ depends on these past prices only through the price at the end of day n. Such an assumption defines a Markov chain, a type of stochastic process that will be studied in this chapter, and which we now formally define.

Let $\{X_n, n = 0, 1, 2, \ldots\}$ be a stochastic process that takes on a finite or countable number of possible values. Unless otherwise mentioned, this set of possible values of the process will be denoted by the set of nonnegative integers $\{0, 1, 2, \ldots\}$. If $X_n = i$, then the process is said to be in state i at time n. We suppose that whenever the process is in state i, there is a fixed probability P_{ij} that it will next be in state j. That is, we suppose that

$$P\{X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}, \ldots, X_1 = i_1, X_0 = i_0\} = P_{ij} \tag{4.1}$$

for all states $i_0, i_1, \ldots, i_{n-1}, i, j$ and all $n \geqslant 0$. Such a stochastic process is known as a *Markov chain*. Eq. (4.1) may be interpreted as stating that, for a Markov chain, the conditional distribution of any future state X_{n+1}, given the past states $X_0, X_1, \ldots, X_{n-1}$ and the present state X_n, is independent of the past states and depends only on the present state.

The value P_{ij} represents the probability that the process will, when in state i, next make a transition into state j. Since probabilities are nonnegative and since the process must make a transition into some state, we have

$$P_{ij} \geqslant 0, \qquad i, j \geqslant 0; \qquad \sum_{j=0}^{\infty} P_{ij} = 1, \qquad i = 0, 1, \ldots$$

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Let $\mathbf{P}$ denote the matrix of one-step transition probabilities P_{ij}, so that

$$\mathbf{P} = \begin{Vmatrix} P_{00} & P_{01} & P_{02} & \cdots \\ P_{10} & P_{11} & P_{12} & \cdots \\ \vdots & \vdots & \vdots & \\ P_{i0} & P_{i1} & P_{i2} & \cdots \\ \vdots & \vdots & \vdots & \end{Vmatrix}$$

Example 4.1 (Forecasting the Weather)**.** Suppose that the chance of rain tomorrow depends on previous weather conditions only through whether or not it is raining today and not on past weather conditions. Suppose also that if it rains today, then it will rain tomorrow with probability α; and if it does not rain today, then it will rain tomorrow with probability β.

If we say that the process is in state 0 when it rains and state 1 when it does not rain, then the preceding is a two-state Markov chain whose transition probabilities are given by

$$\mathbf{P} = \begin{Vmatrix} \alpha & 1-\alpha \\ \beta & 1-\beta \end{Vmatrix}$$ ■

Example 4.2 (A Communications System)**.** Consider a communications system that transmits the digits 0 and 1. Each digit transmitted must pass through several stages, at each of which there is a probability p that the digit entered will be unchanged when it leaves. Letting X_n denote the digit entering the nth stage, then $\{X_n, n = 0, 1, \ldots\}$ is a two-state Markov chain having a transition probability matrix

$$\mathbf{P} = \begin{Vmatrix} p & 1-p \\ 1-p & p \end{Vmatrix}$$ ■

Example 4.3. On any given day Gary is either cheerful (C), so-so (S), or glum (G). If he is cheerful today, then he will be C, S, or G tomorrow with respective probabilities 0.5, 0.4, 0.1. If he is feeling so-so today, then he will be C, S, or G tomorrow with probabilities 0.3, 0.4, 0.3. If he is glum today, then he will be C, S, or G tomorrow with probabilities 0.2, 0.3, 0.5.

Letting X_n denote Gary's mood on the nth day, then $\{X_n, n \geqslant 0\}$ is a three-state Markov chain (state $0 = C$, state $1 = S$, state $2 = G$) with transition probability matrix

$$\mathbf{P} = \begin{Vmatrix} 0.5 & 0.4 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{Vmatrix}$$ ■

Example 4.4 (Transforming a Process into a Markov Chain)**.** Suppose that whether or not it rains today depends on previous weather conditions through the last two days. Specifically, suppose that if it has rained for the past two days, then it will rain tomorrow with probability 0.7; if it rained today but not yesterday, then it will rain

tomorrow with probability 0.5; if it rained yesterday but not today, then it will rain tomorrow with probability 0.4; if it has not rained in the past two days, then it will rain tomorrow with probability 0.2.

If we let the state at time n depend only on whether or not it is raining at time n, then the preceding model is not a Markov chain (why not?). However, we can transform this model into a Markov chain by saying that the state at any time is determined by the weather conditions during both that day and the previous day. In other words, we can say that the process is in

state 0 if it rained both today and yesterday,
state 1 if it rained today but not yesterday,
state 2 if it rained yesterday but not today,
state 3 if it did not rain either yesterday or today.

The preceding would then represent a four-state Markov chain having a transition probability matrix

$$\mathbf{P} = \begin{Vmatrix} 0.7 & 0 & 0.3 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.4 & 0 & 0.6 \\ 0 & 0.2 & 0 & 0.8 \end{Vmatrix}$$

You should carefully check the matrix $\mathbf{P}$, and make sure you understand how it was obtained. ■

Example 4.5 (A Random Walk Model)**.** A Markov chain whose state space is given by the integers $i = 0, \pm 1, \pm 2, \ldots$ is said to be a *random walk* if, for some number $0 < p < 1$,

$$P_{i,i+1} = p = 1 - P_{i,i-1}, \qquad i = 0, \pm 1, \ldots$$

The preceding Markov chain is called a random walk for we may think of it as being a model for an individual walking on a straight line who at each point of time either takes one step to the right with probability p or one step to the left with probability $1 - p$. ■

Example 4.6 (A Gambling Model)**.** Consider a gambler who, at each play of the game, either wins \$1 with probability p or loses \$1 with probability $1 - p$. If we suppose that our gambler quits playing either when he goes broke or he attains a fortune of $\$N$, then the gambler's fortune is a Markov chain having transition probabilities

$$P_{i,i+1} = p = 1 - P_{i,i-1}, \qquad i = 1, 2, \ldots, N-1,$$
$$P_{00} = P_{NN} = 1$$

States 0 and N are called *absorbing* states since once entered they are never left. Note that the preceding is a finite state random walk with absorbing barriers (states 0 and N). ■

Example 4.7. In most of Europe and Asia annual automobile insurance premiums are determined by use of a Bonus Malus (Latin for Good-Bad) system. Each policyholder is given a positive integer valued state and the annual premium is a function of this state (along, of course, with the type of car being insured and the level of insurance). A policyholder's state changes from year to year in response to the number of claims made by that policyholder. Because lower numbered states correspond to lower annual premiums, a policyholder's state will usually decrease if he or she had no claims in the preceding year, and will generally increase if he or she had at least one claim. (Thus, no claims is good and typically results in a decreased premium, while claims are bad and typically result in a higher premium.)

For a given Bonus Malus system, let $s_i(k)$ denote the next state of a policyholder who was in state i in the previous year and who made a total of k claims in that year. If we suppose that the number of yearly claims made by a particular policyholder is a Poisson random variable with parameter λ, then the successive states of this policyholder will constitute a Markov chain with transition probabilities

$$P_{i,j} = \sum_{k:s_i(k)=j} e^{-\lambda}\frac{\lambda^k}{k!}, \quad j \geqslant 0$$

Whereas there are usually many states (20 or so is not atypical), the following table specifies a hypothetical Bonus Malus system having four states.

State	Annual Premium	Next state if			
		0 claims	*1 claim*	*2 claims*	*$\geqslant$ 3 claims*
1	200	1	2	3	4
2	250	1	3	4	4
3	400	2	4	4	4
4	600	3	4	4	4

Thus, for instance, the table indicates that $s_2(0) = 1; s_2(1) = 3; s_2(k) = 4, k \geqslant 2$. Consider a policyholder whose annual number of claims is a Poisson random variable with parameter λ. If a_k is the probability that such a policyholder makes k claims in a year, then

$$a_k = e^{-\lambda}\frac{\lambda^k}{k!}, \quad k \geqslant 0$$

For the Bonus Malus system specified in the preceding table, the transition probability matrix of the successive states of this policyholder is

$$\mathbf{P} = \begin{Vmatrix} a_0 & a_1 & a_2 & 1-a_0-a_1-a_2 \\ a_0 & 0 & a_1 & 1-a_0-a_1 \\ 0 & a_0 & 0 & 1-a_0 \\ 0 & 0 & a_0 & 1-a_0 \end{Vmatrix}$$

■

4.2 Chapman–Kolmogorov Equations

We have already defined the one-step transition probabilities P_{ij}. We now define the n-step transition probabilities P_{ij}^n to be the probability that a process in state i will be in state j after n additional transitions. That is,

$$P_{ij}^n = P\{X_{n+k} = j | X_k = i\}, \quad n \geqslant 0,\ i, j \geqslant 0$$

Of course $P_{ij}^1 = P_{ij}$. The *Chapman–Kolmogorov equations* provide a method for computing these n-step transition probabilities. These equations are

$$P_{ij}^{n+m} = \sum_{k=0}^{\infty} P_{ik}^n P_{kj}^m \quad \text{for all } n,\ m \geqslant 0,\ \text{all } i, j \tag{4.2}$$

and are most easily understood by noting that $P_{ik}^n P_{kj}^m$ represents the probability that starting in i the process will go to state j in $n + m$ transitions through a path which takes it into state k at the nth transition. Hence, summing over all intermediate states k yields the probability that the process will be in state j after $n + m$ transitions. Formally, we have

$$\begin{aligned} P_{ij}^{n+m} &= P\{X_{n+m} = j | X_0 = i\} \\ &= \sum_{k=0}^{\infty} P\{X_{n+m} = j, X_n = k | X_0 = i\} \\ &= \sum_{k=0}^{\infty} P\{X_{n+m} = j | X_n = k, X_0 = i\} P\{X_n = k | X_0 = i\} \\ &= \sum_{k=0}^{\infty} P_{kj}^m P_{ik}^n \end{aligned}$$

If we let $\mathbf{P}^{(n)}$ denote the matrix of n-step transition probabilities P_{ij}^n, then Eq. (4.2) asserts that

$$\mathbf{P}^{(n+m)} = \mathbf{P}^{(n)} \cdot \mathbf{P}^{(m)}$$

where the dot represents matrix multiplication.[1] Hence, in particular,

$$\mathbf{P}^{(2)} = \mathbf{P}^{(1+1)} = \mathbf{P} \cdot \mathbf{P} = \mathbf{P}^2$$

and by induction

$$\mathbf{P}^{(n)} = \mathbf{P}^{(n-1+1)} = \mathbf{P}^{n-1} \cdot \mathbf{P} = \mathbf{P}^n$$

[1] If $\mathbf{A}$ is an $N \times M$ matrix whose element in the ith row and jth column is a_{ij} and $\mathbf{B}$ is an $M \times K$ matrix whose element in the ith row and jth column is b_{ij}, then $\mathbf{A} \cdot \mathbf{B}$ is defined to be the $N \times K$ matrix whose element in the ith row and jth column is $\sum_{k=1}^{M} a_{ik} b_{kj}$.

That is, the n-step transition matrix may be obtained by multiplying the matrix $\mathbf{P}$ by itself n times.

Example 4.8. Consider Example 4.1 in which the weather is considered as a two-state Markov chain. If $\alpha = 0.7$ and $\beta = 0.4$, then calculate the probability that it will rain four days from today given that it is raining today.

Solution: The one-step transition probability matrix is given by

$$\mathbf{P} = \begin{Vmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{Vmatrix}$$

Hence,

$$\begin{aligned} \mathbf{P}^{(2)} = \mathbf{P}^2 &= \begin{Vmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{Vmatrix} \cdot \begin{Vmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{Vmatrix} \\ &= \begin{Vmatrix} 0.61 & 0.39 \\ 0.52 & 0.48 \end{Vmatrix}, \\ \mathbf{P}^{(4)} = (\mathbf{P}^2)^2 &= \begin{Vmatrix} 0.61 & 0.39 \\ 0.52 & 0.48 \end{Vmatrix} \cdot \begin{Vmatrix} 0.61 & 0.39 \\ 0.52 & 0.48 \end{Vmatrix} \\ &= \begin{Vmatrix} 0.5749 & 0.4251 \\ 0.5668 & 0.4332 \end{Vmatrix} \end{aligned}$$

and the desired probability P_{00}^4 equals 0.5749. ■

Example 4.9. Consider Example 4.4. Given that it rained on Monday and Tuesday, what is the probability that it will rain on Thursday?

Solution: The two-step transition matrix is given by

$$\begin{aligned} \mathbf{P}^{(2)} = \mathbf{P}^2 &= \begin{Vmatrix} 0.7 & 0 & 0.3 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.4 & 0 & 0.6 \\ 0 & 0.2 & 0 & 0.8 \end{Vmatrix} \cdot \begin{Vmatrix} 0.7 & 0 & 0.3 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.4 & 0 & 0.6 \\ 0 & 0.2 & 0 & 0.8 \end{Vmatrix} \\ &= \begin{Vmatrix} 0.49 & 0.12 & 0.21 & 0.18 \\ 0.35 & 0.20 & 0.15 & 0.30 \\ 0.20 & 0.12 & 0.20 & 0.48 \\ 0.10 & 0.16 & 0.10 & 0.64 \end{Vmatrix} \end{aligned}$$

Since rain on Thursday is equivalent to the process being in either state 0 or state 1 on Thursday, the desired probability is given by $P_{00}^2 + P_{01}^2 = 0.49 + 0.12 = 0.61$. ■

Example 4.10. An urn always contains 2 balls. Ball colors are red and blue. At each stage a ball is randomly chosen and then replaced by a new ball, which with probability 0.8 is the same color, and with probability 0.2 is the opposite color, as the ball it replaces. If initially both balls are red, find the probability that the fifth ball selected is red.

Solution: To find the desired probability we first define an appropriate Markov chain. This can be accomplished by noting that the probability that a selection is red is determined by the composition of the urn at the time of the selection. So, let us define X_n to be the number of red balls in the urn after the nth selection and subsequent replacement. Then $X_n, n \geq 0$, is a Markov chain with states 0, 1, 2 and with transition probability matrix $\mathbf{P}$ given by

$$\begin{pmatrix} 0.8 & 0.2 & 0 \\ 0.1 & 0.8 & 0.1 \\ 0 & 0.2 & 0.8 \end{pmatrix}$$

To understand the preceding, consider for instance $P_{1,0}$. Now, to go from 1 red ball in the urn to 0 red balls, the ball chosen must be red (which occurs with probability 0.5) and it must then be replaced by a ball of opposite color (which occurs with probability 0.2), showing that

$$P_{1,0} = (0.5)(0.2) = 0.1$$

To determine the probability that the fifth selection is red, condition on the number of red balls in the urn after the fourth selection. This yields

$$\begin{aligned} &P(\text{fifth selection is red}) \\ &\quad = \sum_{i=0}^{2} P(\text{fifth selection is red} | X_4 = i) P(X_4 = i | X_0 = 2) \\ &\quad = (0) P_{2,0}^4 + (0.5) P_{2,1}^4 + (1) P_{2,2}^4 \\ &\quad = 0.5 P_{2,1}^4 + P_{2,2}^4 \end{aligned}$$

To calculate the preceding we compute $\mathbf{P}^4$. Doing so yields

$$P_{2,1}^4 = 0.4352, \quad P_{2,2}^4 = 0.4872$$

giving the answer $P(\text{fifth selection is red}) = 0.7048$. ■

Example 4.11. Suppose that balls are successively distributed among 8 urns, with each ball being equally likely to be put in any of these urns. What is the probability that there will be exactly 3 nonempty urns after 9 balls have been distributed?

Solution: If we let X_n be the number of nonempty urns after n balls have been distributed, then $X_n, n \geqslant 0$ is a Markov chain with states $0, 1, \ldots, 8$ and transition probabilities

$$P_{i,i} = i/8 = 1 - P_{i,i+1}, \quad i = 0, 1, \ldots, 8$$

The desired probability is $P_{0,3}^9 = P_{1,3}^8$, where the equality follows because $P_{0,1} = 1$. Now, starting with 1 occupied urn, if we had wanted to determine the entire probability distribution of the number of occupied urns after 8 additional balls

had been distributed we would need to consider the transition probability matrix with states $1, 2, \ldots, 8$. However, because we only require the probability, starting with a single occupied urn, that there are 3 occupied urns after an additional 8 balls have been distributed we can make use of the fact that the state of the Markov chain cannot decrease to collapse all states $4, 5, \ldots, 8$ into a single state 4 with the interpretation that the state is 4 whenever four or more of the urns are occupied. Consequently, we need only determine the eight-step transition probability $P_{1,3}^8$ of the Markov chain with states 1, 2, 3, 4 having transition probability matrix $\mathbf{P}$ given by

$$\begin{pmatrix} 1/8 & 7/8 & 0 & 0 \\ 0 & 2/8 & 6/8 & 0 \\ 0 & 0 & 3/8 & 5/8 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Raising the preceding matrix to the power 4 yields the matrix $\mathbf{P}^4$ given by

$$\begin{pmatrix} 0.0002 & 0.0256 & 0.2563 & 0.7178 \\ 0 & 0.0039 & 0.0952 & 0.9009 \\ 0 & 0 & 0.0198 & 0.9802 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Hence,

$$\begin{aligned} P_{1,3}^8 &= 0.0002 \times 0.2563 + 0.0256 \times 0.0952 + 0.2563 \times 0.0198 \\ &\quad + 0.7178 \times 0 = 0.00756 \end{aligned}$$

■

Consider a Markov chain with transition probabilities P_{ij}. Let $\mathscr{A}$ be a set of states, and suppose we are interested in the probability that the Markov chain ever enters any of the states in $\mathscr{A}$ by time m. That is, for a given state $i \notin \mathscr{A}$, we are interested in determining

$$\beta = P(X_k \in \mathscr{A} \text{ for some } k = 1, \ldots, m | X_0 = i)$$

To determine the preceding probability we will define a Markov chain $\{W_n, n \geqslant 0\}$ whose states are the states that are not in $\mathscr{A}$ plus an additional state, which we will call A in our general discussion (though in specific examples we will usually give it a different name). Once the $\{W_n\}$ Markov chain enters state A it remains there forever.

The new Markov chain is defined as follows. Letting X_n denote the state at time n of the Markov chain with transition probabilities $P_{i,j}$, define

$$N = \min\{n : X_n \in \mathscr{A}\}$$

and let $N = \infty$ if $X_n \notin \mathscr{A}$ for all n. In words, N is the first time the Markov chain enters the set of states $\mathscr{A}$. Now, define

$$W_n = \begin{cases} X_n, & \text{if } n < N \\ A, & \text{if } n \geqslant N \end{cases}$$

So the state of the $\{W_n\}$ process is equal to the state of the original Markov chain up to the point when the original Markov chain enters a state in $\mathscr{A}$. At that time the new process goes to state A and remains there forever. From this description it follows that $W_n, n \geqslant 0$ is a Markov chain with states $i, i \notin \mathscr{A}$, A and with transition probabilities $Q_{i,j}$, given by

$$
\begin{aligned}
Q_{i,j} &= P_{i,j}, \quad \text{if} \quad i \notin \mathscr{A}, j \notin \mathscr{A} \\
Q_{i,A} &= \sum_{j \in \mathscr{A}} P_{i,j}, \quad \text{if} \quad i \notin \mathscr{A} \\
Q_{A,A} &= 1
\end{aligned}
$$

Because the original Markov chain will have entered a state in $\mathscr{A}$ by time m if and only if the state at time m of the new Markov chain is A, we see that

$$
\begin{aligned}
&P(X_k \in \mathscr{A} \text{ for some } k = 1, \ldots, m | X_0 = i) \\
&\quad = P(W_m = A | X_0 = i) = P(W_m = A | W_0 = i) = Q_{i,A}^m
\end{aligned}
$$

That is, the desired probability is equal to an m-step transition probability of the new chain.

Example 4.12. In a sequence of independent flips of a fair coin, let N denote the number of flips until there is a run of three consecutive heads. Find

(a) $P(N \leq 8)$ and
(b) $P(N = 8)$.

Solution: To determine $P(N \leq 8)$, define a Markov chain with states $0, 1, 2, 3$ where for $i < 3$ state i means that we currently are on a run of i consecutive heads, and where state 3 means that a run of three consecutive heads has already occurred. Thus, the transition probability matrix is

$$
\mathbf{P} = \begin{pmatrix}
1/2 & 1/2 & 0 & 0 \\
1/2 & 0 & 1/2 & 0 \\
1/2 & 0 & 0 & 1/2 \\
0 & 0 & 0 & 1
\end{pmatrix}
$$

where, for instance, the values for row 2 are obtained by noting that if we currently are on a run of size 1 then the next state will be 0 if the next flip is a tail, or 2 if it is a head. Hence, $P_{1,0} = P_{1,2} = 1/2$. Because there would be a run of three consecutive heads within the first eight flips if and only if $X_8 = 3$, the desired probability is $P_{0,3}^8$. Squaring $\mathbf{P}$ to obtain $\mathbf{P}^2$, then squaring the result to obtain $\mathbf{P}^4$, and then squaring that matrix gives the result

$$
\mathbf{P}^8 = \begin{pmatrix}
81/256 & 44/256 & 24/256 & 107/256 \\
68/256 & 37/256 & 20/256 & 131/256 \\
44/256 & 24/256 & 13/256 & 175/256 \\
0 & 0 & 0 & 1
\end{pmatrix}
$$

Hence, the probability that there will be a run of three consecutive heads within the first eight flips is $107/256 \approx .4180$.
(b) Noting that $N = 8$ if the pattern hasn't yet occurred in the first 7 transitions, the state after 7 transitions is 2, and the next flip lands heads, shows that

$$P(N = 8) = \frac{1}{2} P_{0,2}^7.$$ ■

We can also use Markov chains to determine probabilities concerning the time until a pattern appears when the data itself comes from a Markov chain. We illustrate this by an example.

Example 4.13. Let $\{X_n, n \geq 0\}$ be a Markov chain with states 0, 1, 2, 3 and transition probabilities $P_{i,j}$, $i, j = 0, 1, 2, 3$, and let N denote the number of transitions, starting in state 0, until the pattern 1, 2, 1, 2 appears. That is,

$$N = \min\{n \geq 4 : X_{n-3} = 1, X_{n-2} = 2, X_{n-1} = 1, X_n = 2\}.$$

Suppose we are interested in evaluating $P(N \leq k)$ for some specified value k. To do so, we define a new Markov chain $\{Y_n, n \geq 0\}$, that tracks the progress towards the pattern. The Y_n are defined as follows:

- If the pattern has appeared by the nth transition—that is, if $X_0, \ldots, X_n$ includes 1, 2, 1, 2—then $Y_n = 4$.
- If the pattern has not yet appeared by the nth transition

  $Y_n = 1$ if $X_n = 1$ and $(X_{n-2}, X_{n-1}) \neq (1, 2)$.
  $Y_n = 2$ if $X_{n-1} = 1$, $X_n = 2$.
  $Y_n = 3$ if $X_{n-2} = 1$, $X_{n-1} = 2$, $X_n = 1$.
  $Y_n = 5$ if $X_n = 2$, $X_{n-1} \neq 1$.
  $Y_n = 6$ if $X_n = 0$.
  $Y_n = 7$ if $X_n = 3$

Thus, for $i = 1, 2, 3, 4$, $Y_n = i$ signifies that we are i steps into the pattern (or in the case $i = 4$ that the pattern has appeared). $Y_n = 5$ (or 6 or 7) if there is no current progress with regards to the pattern and the current state is 2 (or 0 or 3). The desired probability $P(N \leq k)$ is equal to the probability that the number of transitions of the Markov chain $\{Y_n\}$ to go from state 6 to state 4 is less than or equal to k. Because state 4 is an absorbing state of this chain, this probability is $Q_{6,4}^k$ where $Q_{i,j}$ are the transition probabilities of the Markov chain $\{Y_n\}$. ■

Suppose now that we want to compute the probability that the $\{X_n, n \geqslant 0\}$ chain, starting in state i, enters state j at time m without ever entering any of the states in $\mathscr{A}$, where neither i nor j is in $\mathscr{A}$. That is, for $i, j \notin \mathscr{A}$, we are interested in

$$\alpha = P(X_m = j, X_k \notin \mathscr{A}, k = 1, \ldots, m-1 | X_0 = i)$$

Noting that the event that $X_m = j, X_k \notin \mathscr{A}, k = 1, \ldots, m-1$ is equivalent to the event that $W_m = j$, it follows that for $i, j \notin \mathscr{A}$,

$$P(X_m = j, X_k \notin \mathscr{A}, k = 1, \ldots, m-1 | X_0 = i) = P(W_m = j | X_0 = i)$$
$$= P(W_m = j | W_0 = i) = Q^m_{i,j}.$$

Example 4.14. Consider a Markov chain with states 1, 2, 3, 4, 5, and suppose that we want to compute

$$P(X_4 = 2, X_3 \leqslant 2, X_2 \leqslant 2, X_1 \leqslant 2 | X_0 = 1)$$

That is, we want the probability that, starting in state 1, the chain is in state 2 at time 4 and has never entered any of the states in the set $\mathscr{A} = \{3, 4, 5\}$.

To compute this probability all we need to know are the transition probabilities $P_{11}, P_{12}, P_{21}, P_{22}$. So, suppose that

$$P_{11} = 0.3 \quad P_{12} = 0.3$$
$$P_{21} = 0.1 \quad P_{22} = 0.2$$

Then we consider the Markov chain having states 1, 2, 3 (we are giving state A the name 3), and having the transition probability matrix $\mathbf{Q}$ as follows:

$$\begin{pmatrix} 0.3 & 0.3 & 0.4 \\ 0.1 & 0.2 & 0.7 \\ 0 & 0 & 1 \end{pmatrix}$$

The desired probability is Q^4_{12}. Raising $\mathbf{Q}$ to the power 4 yields the matrix

$$\begin{pmatrix} 0.0219 & 0.0285 & 0.9496 \\ 0.0095 & 0.0124 & 0.9781 \\ 0 & 0 & 1 \end{pmatrix}$$

Hence, the desired probability is $\alpha = 0.0285$. ■

When $i \notin \mathscr{A}$ but $j \in \mathscr{A}$ we can determine the probability

$$\alpha = P(X_m = j, X_k \notin \mathscr{A}, k = 1, \ldots, m-1 | X_0 = i)$$

as follows.

$$\alpha = \sum_{r \notin \mathscr{A}} P(X_m = j, X_{m-1} = r, X_k \notin \mathscr{A}, k = 1, \ldots, m-2 | X_0 = i)$$
$$= \sum_{r \notin \mathscr{A}} P(X_m = j | X_{m-1} = r, X_k \notin \mathscr{A}, k = 1, \ldots, m-2, X_0 = i)$$
$$\times P(X_{m-1} = r, X_k \notin \mathscr{A}, k = 1, \ldots, m-2 | X_0 = i)$$
$$= \sum_{r \notin \mathscr{A}} P_{r,j} \, P(X_{m-1} = r, X_k \notin \mathscr{A}, k = 1, \ldots, m-2 | X_0 = i)$$
$$= \sum_{r \notin \mathscr{A}} P_{r,j} \, Q^{m-1}_{i,r}$$

Also, when $i \in \mathscr{A}$ we could determine

$$\alpha = P(X_m = j, X_k \notin \mathscr{A}, k = 1, \ldots, m-1 | X_0 = i)$$

by conditioning on the first transition to obtain

$$\begin{aligned}\alpha &= \sum_{r \notin \mathscr{A}} P(X_m = j, X_k \notin \mathscr{A}, \\ &\qquad k = 1, \ldots, m-1 | X_0 = i, X_1 = r) P(X_1 = r | X_0 = i) \\ &= \sum_{r \notin \mathscr{A}} P(X_{m-1} = j, X_k \notin \mathscr{A}, k = 1, \ldots, m-2 | X_0 = r) P_{i,r}\end{aligned}$$

For instance, if $i \in \mathscr{A}$, $j \notin \mathscr{A}$ then the preceding equation yields

$$P(X_m = j, X_k \notin \mathscr{A}, k = 1, \ldots, m-1 | X_0 = i) = \sum_{r \notin \mathscr{A}} Q_{r,j}^{m-1} P_{i,r}$$

We can also compute the conditional probability of X_n given that the chain starts in state i and has not entered any state in $\mathscr{A}$ by time n, as follows. For $i, j \notin \mathscr{A}$,

$$\begin{aligned}&P\{X_n = j | X_0 = i, X_k \notin \mathscr{A}, k = 1, \ldots, n\} \\ &\quad = \frac{P\{X_n = j, X_k \notin \mathscr{A}, k = 1, \ldots, n | X_0 = i\}}{P\{X_k \notin \mathscr{A}, k = 1, \ldots, n | X_0 = i\}} = \frac{Q_{i,j}^n}{\sum_{r \notin \mathscr{A}} Q_{i,r}^n}\end{aligned}$$

Remark. So far, all of the probabilities we have considered are conditional probabilities. For instance, P_{ij}^n is the probability that the state at time n is j given that the initial state at time 0 is i. If the unconditional distribution of the state at time n is desired, it is necessary to specify the probability distribution of the initial state. Let us denote this by

$$\alpha_i \equiv P\{X_0 = i\}, \quad i \geqslant 0 \left(\sum_{i=0}^{\infty} \alpha_i = 1\right)$$

All unconditional probabilities may be computed by conditioning on the initial state. That is,

$$\begin{aligned}P\{X_n = j\} &= \sum_{i=0}^{\infty} P\{X_n = j | X_0 = i\} P\{X_0 = i\} \\ &= \sum_{i=0}^{\infty} P_{ij}^n \alpha_i\end{aligned}$$

For instance, if $\alpha_0 = 0.4, \alpha_1 = 0.6$, in Example 4.8, then the (unconditional) probability that it will rain four days after we begin keeping weather records is

$$P\{X_4 = 0\} = 0.4 P_{00}^4 + 0.6 P_{10}^4$$

$$= (0.4)(0.5749) + (0.6)(0.5668)$$
$$= 0.5700$$

4.3 Classification of States

State j is said to be *accessible* from state i if $P_{ij}^n > 0$ for some $n \geqslant 0$. Note that this implies that state j is accessible from state i if and only if, starting in i, it is possible that the process will ever enter state j. This is true since if j is not accessible from i, then

$$\begin{aligned} P\{\text{ever be in } j | \text{start in } i\} &= P\left\{ \bigcup_{n=0}^{\infty} \{X_n = j\} \,\middle|\, X_0 = i \right\} \\ &\leq \sum_{n=0}^{\infty} P\{X_n = j | X_0 = i\} \\ &= \sum_{n=0}^{\infty} P_{ij}^n \\ &= 0 \end{aligned}$$

Two states i and j that are accessible to each other are said to *communicate*, and we write $i \leftrightarrow j$.

Note that any state communicates with itself since, by definition,

$$P_{ii}^0 = P\{X_0 = i | X_0 = i\} = 1$$

The relation of communication satisfies the following three properties:

- **(i)** State i communicates with state i, all $i \geqslant 0$.
- **(ii)** If state i communicates with state j, then state j communicates with state i.
- **(iii)** If state i communicates with state j, and state j communicates with state k, then state i communicates with state k.

Properties (i) and (ii) follow immediately from the definition of communication. To prove (iii) suppose that i communicates with j, and j communicates with k. Thus, there exist integers n and m such that $P_{ij}^n > 0, P_{jk}^m > 0$. Now by the Chapman–Kolmogorov equations, we have

$$P_{ik}^{n+m} = \sum_{r=0}^{\infty} P_{ir}^n P_{rk}^m \geqslant P_{ij}^n P_{jk}^m > 0$$

Hence, state k is accessible from state i. Similarly, we can show that state i is accessible from state k. Hence, states i and k communicate.

Two states that communicate are said to be in the same *class*. It is an easy consequence of (i), (ii), and (iii) that any two classes of states are either identical or disjoint.

In other words, the concept of communication divides the state space up into a number of separate classes. The Markov chain is said to be *irreducible* if there is only one class, that is, if all states communicate with each other.

Example 4.15. Consider the Markov chain consisting of the three states 0, 1, 2 and having transition probability matrix

$$\mathbf{P} = \begin{Vmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & \frac{1}{3} & \frac{2}{3} \end{Vmatrix}$$

It is easy to verify that this Markov chain is irreducible. For example, it is possible to go from state 0 to state 2 since

$$0 \to 1 \to 2$$

That is, one way of getting from state 0 to state 2 is to go from state 0 to state 1 (with probability $\frac{1}{2}$) and then go from state 1 to state 2 (with probability $\frac{1}{4}$). ■

Example 4.16. Consider a Markov chain consisting of the four states 0, 1, 2, 3 and having transition probability matrix

$$\mathbf{P} = \begin{Vmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & 0 & 1 \end{Vmatrix}$$

The classes of this Markov chain are $\{0, 1\}$, $\{2\}$, and $\{3\}$. Note that while state 0 (or 1) is accessible from state 2, the reverse is not true. Since state 3 is an absorbing state, that is, $P_{33} = 1$, no other state is accessible from it. ■

For any state i we let f_i denote the probability that, starting in state i, the process will ever reenter state i. State i is said to be *recurrent* if $f_i = 1$ and *transient* if $f_i < 1$.

Suppose that the process starts in state i and i is recurrent. Hence, with probability 1, the process will eventually reenter state i. However, by the definition of a Markov chain, it follows that the process will be starting over again when it reenters state i and, therefore, state i will eventually be visited again. Continual repetition of this argument leads to the conclusion that *if state i is recurrent then, starting in state i, the process will reenter state i again and again and again—in fact, infinitely often.*

On the other hand, suppose that state i is transient. Hence, each time the process enters state i there will be a positive probability, namely, $1 - f_i$, that it will never again enter that state. Therefore, starting in state i, the probability that the process will be in state i for exactly n time periods equals $f_i^{n-1}(1 - f_i)$, $n \geqslant 1$. In other words, *if state i is transient then, starting in state i, the number of time periods that the process will be in state i has a geometric distribution with finite mean* $1/(1 - f_i)$.

From the preceding two paragraphs, it follows that *state i is recurrent if and only if, starting in state i, the expected number of time periods that the process is in state i is infinite*. But, letting

$$I_n = \begin{cases} 1, & \text{if } X_n = i \\ 0, & \text{if } X_n \neq i \end{cases}$$

we have that $\sum_{n=0}^{\infty} I_n$ represents the number of periods that the process is in state i. Also,

$$\begin{aligned} E\left[\sum_{n=0}^{\infty} I_n | X_0 = i\right] &= \sum_{n=0}^{\infty} E[I_n | X_0 = i] \\ &= \sum_{n=0}^{\infty} P\{X_n = i | X_0 = i\} \\ &= \sum_{n=0}^{\infty} P_{ii}^n \end{aligned}$$

We have thus proven the following.

Proposition 4.1. *State i is*

$$\text{recurrent if } \sum_{n=1}^{\infty} P_{ii}^n = \infty,$$

$$\text{transient if } \sum_{n=1}^{\infty} P_{ii}^n < \infty$$

The argument leading to the preceding proposition is doubly important because it also shows that a transient state will only be visited a finite number of times (hence the name transient). This leads to the conclusion that in a finite-state Markov chain not all states can be transient. To see this, suppose the states are $0, 1, \ldots, M$ and suppose that they are all transient. Then after a finite amount of time (say, after time T_0) state 0 will never be visited, and after a time (say, T_1) state 1 will never be visited, and after a time (say, T_2) state 2 will never be visited, and so on. Thus, after a finite time $T = \max\{T_0, T_1, \ldots, T_M\}$ no states will be visited. But as the process must be in some state after time T we arrive at a contradiction, which shows that at least one of the states must be recurrent.

Another use of Proposition 4.1 is that it enables us to show that recurrence is a class property.

Corollary 4.2. *If state i is recurrent, and state i communicates with state j, then state j is recurrent.*

Proof. To prove this we first note that, since state i communicates with state j, there exist integers k and m such that $P_{ij}^k > 0$, $P_{ji}^m > 0$. Now, for any integer n

$$P_{jj}^{m+n+k} \geqslant P_{ji}^m P_{ii}^n P_{ij}^k$$

This follows since the left side of the preceding is the probability of going from j to j in $m+n+k$ steps, while the right side is the probability of going from j to j in $m+n+k$ steps via a path that goes from j to i in m steps, then from i to i in an additional n steps, then from i to j in an additional k steps.

From the preceding we obtain, by summing over n, that

$$\sum_{n=1}^{\infty} P_{jj}^{m+n+k} \geqslant P_{ji}^m P_{ij}^k \sum_{n=1}^{\infty} P_{ii}^n = \infty$$

since $P_{ji}^m P_{ij}^k > 0$ and $\sum_{n=1}^{\infty} P_{ii}^n$ is infinite since state i is recurrent. Thus, by Proposition 4.1 it follows that state j is also recurrent. ■

Remarks. **(i)** Corollary 4.2 also implies that transience is a class property. For if state i is transient and communicates with state j, then state j must also be transient. For if j were recurrent then, by Corollary 4.2, i would also be recurrent and hence could not be transient.

(ii) Corollary 4.2 along with our previous result that not all states in a finite Markov chain can be transient leads to the conclusion that all states of a finite irreducible Markov chain are recurrent.

Example 4.17. Let the Markov chain consisting of the states 0, 1, 2, 3 have the transition probability matrix

$$\mathbf{P} = \begin{Vmatrix} 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{Vmatrix}$$

Determine which states are transient and which are recurrent.

Solution: It is a simple matter to check that all states communicate and, hence, since this is a finite chain, all states must be recurrent. ■

Example 4.18. Consider the Markov chain having states 0, 1, 2, 3, 4 and

$$\mathbf{P} = \begin{Vmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{4} & \frac{1}{4} & 0 & 0 & \frac{1}{2} \end{Vmatrix}$$

Determine the recurrent state.

Solution: This chain consists of the three classes $\{0, 1\}$, $\{2, 3\}$, and $\{4\}$. The first two classes are recurrent and the third transient. ■

Example 4.19 (A Random Walk)**.** Consider a Markov chain whose state space consists of the integers $i = 0, \pm 1, \pm 2, \ldots$, and has transition probabilities given by

$$P_{i,i+1} = p = 1 - P_{i,i-1}, \quad i = 0, \pm 1, \pm 2, \ldots$$

where $0 < p < 1$. In other words, on each transition the process either moves one step to the right (with probability p) or one step to the left (with probability $1 - p$). One colorful interpretation of this process is that it represents the wanderings of a drunken man as he walks along a straight line. Another is that it represents the winnings of a gambler who on each play of the game either wins or loses one dollar.

Since all states clearly communicate, it follows from Corollary 4.2 that they are either all transient or all recurrent. So let us consider state 0 and attempt to determine if $\sum_{n=1}^{\infty} P_{00}^n$ is finite or infinite.

Since it is impossible to be even (using the gambling model interpretation) after an odd number of plays we must, of course, have that

$$P_{00}^{2n-1} = 0, \quad n = 1, 2, \ldots$$

On the other hand, we would be even after $2n$ trials if and only if we won n of these and lost n of these. Because each play of the game results in a win with probability p and a loss with probability $1 - p$, the desired probability is thus the binomial probability

$$P_{00}^{2n} = \binom{2n}{n} p^n (1-p)^n = \frac{(2n)!}{n!n!} (p(1-p))^n, \quad n = 1, 2, 3, \ldots$$

By using an approximation, due to Stirling, which asserts that

$$n! \sim n^{n+1/2} e^{-n} \sqrt{2\pi} \tag{4.3}$$

where we say that $a_n \sim b_n$ when $\lim_{n\to\infty} a_n/b_n = 1$, we obtain

$$\binom{2n}{n} \sim \frac{(2n)^{2n+1/2}\, e^{-2n}\, \sqrt{2\pi}}{n^{2n+1}\, e^{-2n}\, (2\pi)} = \frac{2^{2n}}{\sqrt{n\pi}}$$

Hence,

$$P_{00}^{2n} \sim \frac{(4p(1-p))^n}{\sqrt{\pi n}}$$

Now it is easy to verify, for positive a_n, b_n, that if $a_n \sim b_n$, then $\sum_n a_n < \infty$ if and only if $\sum_n b_n < \infty$. Hence, $\sum_{n=1}^{\infty} P_{00}^n$ will converge if and only if

$$\sum_{n=1}^{\infty} \frac{(4p(1-p))^n}{\sqrt{\pi n}}$$

does. However, $4p(1-p) \leq 1$ with equality holding if and only if $p = \frac{1}{2}$. Hence, $\sum_{n=1}^{\infty} P_{00}^n = \infty$ if and only if $p = \frac{1}{2}$. Thus, the chain is recurrent when $p = \frac{1}{2}$ and transient if $p \neq \frac{1}{2}$.

When $p = \frac{1}{2}$, the preceding process is called a *symmetric random walk*. We could also look at symmetric random walks in more than one dimension. For instance, in the two-dimensional symmetric random walk the process would, at each transition, either take one step to the left, right, up, or down, each having probability $\frac{1}{4}$. That is, the state is the pair of integers (i, j) and the transition probabilities are given by

$$P_{(i,j),(i+1,j)} = P_{(i,j),(i-1,j)} = P_{(i,j),(i,j+1)} = P_{(i,j),(i,j-1)} = \tfrac{1}{4}$$

By using the same method as in the one-dimensional case, we now show that this Markov chain is also recurrent.

Since the preceding chain is irreducible, it follows that all states will be recurrent if state $\mathbf{0} = (0, 0)$ is recurrent. So consider $P_{\mathbf{00}}^{2n}$. Now after $2n$ steps, the chain will be back in its original location if for some $i, 0 \leqslant i \leqslant n$, the $2n$ steps consist of i steps to the left, i to the right, $n - i$ up, and $n - i$ down. Since each step will be either of these four types with probability $\frac{1}{4}$, it follows that the desired probability is a multinomial probability. That is,

$$\begin{aligned}
P_{\mathbf{00}}^{2n} &= \sum_{i=0}^{n} \frac{(2n)!}{i!i!(n-i)!(n-i)!} \left(\frac{1}{4}\right)^{2n} \\
&= \sum_{i=0}^{n} \frac{(2n)!}{n!n!} \frac{n!}{(n-i)!i!} \frac{n!}{(n-i)!i!} \left(\frac{1}{4}\right)^{2n} \\
&= \left(\frac{1}{4}\right)^{2n} \binom{2n}{n} \sum_{i=0}^{n} \binom{n}{i} \binom{n}{n-i} \\
&= \left(\frac{1}{4}\right)^{2n} \binom{2n}{n} \binom{2n}{n} \qquad (4.4)
\end{aligned}$$

where the last equality uses the combinatorial identity

$$\binom{2n}{n} = \sum_{i=0}^{n} \binom{n}{i} \binom{n}{n-i}$$

which follows upon noting that both sides represent the number of subgroups of size n one can select from a set of n white and n black objects. Now,

$$\begin{aligned}
\binom{2n}{n} &= \frac{(2n)!}{n!n!} \\
&\sim \frac{(2n)^{2n+1/2} e^{-2n} \sqrt{2\pi}}{n^{2n+1} e^{-2n} (2\pi)} \quad \text{by Stirling's approximation}
\end{aligned}$$

$$= \frac{4^n}{\sqrt{\pi n}}$$

Hence, from Eq. (4.4) we see that

$$P_{00}^{2n} \sim \frac{1}{\pi n}$$

which shows that $\sum_n P_{00}^{2n} = \infty$, and thus all states are recurrent.

Interestingly enough, whereas the symmetric random walks in one and two dimensions are both recurrent, all higher-dimensional symmetric random walks turn out to be transient. (For instance, the three-dimensional symmetric random walk is at each transition equally likely to move in any of six ways—either to the left, right, up, down, in, or out.) ■

Remark. For the one-dimensional random walk of Example 4.19 here is a direct argument for establishing recurrence in the symmetric case, and for determining the probability that it ever returns to state 0 in the nonsymmetric case. Let

$$\beta = P\{\text{ever return to } 0\}$$

To determine β, start by conditioning on the initial transition to obtain

$$\beta = P\{\text{ever return to } 0 | X_1 = 1\}p + P\{\text{ever return to } 0 | X_1 = -1\}(1 - p) \quad (4.5)$$

Now, let α denote the probability that the Markov chain will ever return to state 0 given that it is currently in state 1. Because the Markov chain will always increase by 1 with probability p or decrease by 1 with probability $1 - p$ no matter what its current state, note that α is also the probability that the Markov chain currently in state i will ever enter state $i - 1$, for any i. To obtain an equation for α, condition on the next transition to obtain

$$\begin{aligned}\alpha &= P\{\text{ever return} | X_1 = 1, X_2 = 0\}(1 - p) + P\{\text{ever return} | X_1 = 1, X_2 = 2\}p \\ &= 1 - p + P\{\text{ever return} | X_1 = 1, X_2 = 2\}p \\ &= 1 - p + p\alpha^2\end{aligned}$$

where the final equation follows by noting that in order for the chain to ever go from state 2 to state 0 it must first go to state 1—and the probability of that ever happening is α—and if it does eventually go to state 1 then it must still go to state 0—and the conditional probability of that ever happening is also α. Therefore,

$$\alpha = 1 - p + p\alpha^2$$

The two roots of this equation are $\alpha = 1$ and $\alpha = (1 - p)/p$. Consequently, in the case of the symmetric random walk where $p = 1/2$ we can conclude that $\alpha = 1$. By symmetry, the probability that the symmetric random walk will ever enter state 0 given

that it is currently in state -1 is also 1, proving that the symmetric random walk is recurrent.

Suppose now that $p > 1/2$. In this case, it can be shown (see Exercise 17 at the end of this chapter) that $P\{\text{ever return to } 0 | X_1 = -1\} = 1$. Consequently, Eq. (4.5) reduces to

$$\beta = \alpha p + 1 - p$$

Because the random walk is transient in this case we know that $\beta < 1$, showing that $\alpha \neq 1$. Therefore, $\alpha = (1 - p)/p$, yielding that

$$\beta = 2(1 - p), \quad p > 1/2$$

Similarly, when $p < 1/2$ we can show that $\beta = 2p$. Thus, in general

$$P\{\text{ever return to } 0\} = 2\min(p, 1 - p)$$ ■

In our next example we use the recurrence of the symmetric random walk to construct an example where $E[\sum_{n=1}^{\infty} X_n] \neq \sum_{n=1}^{\infty} E[X_n]$.

Example 4.20. Whereas it is true that $E[\sum_{n=1}^{\infty} X_n] = \sum_{n=1}^{\infty} E[X_n]$ when the random variables $X_n, n \geq 1$ are all nonnegative, it is not true in general. For an example where it does not hold, suppose that $Y_1, Y_2, \ldots$ are independent and identically distributed with $P(Y_n = 1) = P(Y_n = -1) = 1/2, n \geq 1$. Note that $E[Y_n] = 0$. Let

$$N = \min(k : Y_1 + \ldots + Y_k = 1)$$

and note that it follows from the fact that the symmetric random walk is recurrent that N will be finite with probability 1. Now, let

$$I_n = \begin{cases} 1, & \text{if } n \leq N \\ 0, & \text{if } n > N \end{cases}$$

Because $I_n = 1$ if $N > n - 1$, and it is equal to 0 otherwise, it follows that the value of I_n is determined by $Y_1, \ldots, Y_{n-1}$. Indeed, because N is defined to be the first time the sum of the Y's is equal to 1, it follows that

$$\{I_n = 1\} = \{N > n - 1\} = \{Y_1 \neq 1, Y_1 + Y_2 \neq 1, \ldots, Y_1 + \ldots + Y_{n-1} \neq 1\}$$

which shows that I_n and Y_n are independent. Now, define $X_n, n \geq 1$, by

$$X_n = Y_n I_n = \begin{cases} Y_n, & \text{if } n \leq N \\ 0, & \text{if } n > N \end{cases}$$

By the independence of Y_n and I_n,

$$E[X_n] = E[Y_n]\, E[I_n] = 0$$

showing that

$$\sum_{n=1}^{\infty} E[X_n] = 0$$

However,

$$\sum_{n=1}^{\infty} X_n = \sum_{n=1}^{\infty} Y_n I_n = \sum_{n=1}^{N} Y_n = 1$$

and so

$$E[\sum_{n=1}^{\infty} X_n] = 1$$

Thus,

$$E[\sum_{n=1}^{\infty} X_n] = 1 \text{ and } \sum_{n=1}^{\infty} E[X_n] = 0$$ ■

Example 4.21 (On the Ultimate Instability of the Aloha Protocol)**.** Consider a communications facility in which the numbers of messages arriving during each of the time periods $n = 1, 2, \ldots$ are independent and identically distributed random variables. Let $a_i = P\{i \text{ arrivals}\}$, and suppose that $a_0 + a_1 < 1$. Each arriving message will transmit at the end of the period in which it arrives. If exactly one message is transmitted, then the transmission is successful and the message leaves the system. However, if at any time two or more messages simultaneously transmit, then a collision is deemed to occur and these messages remain in the system. Once a message is involved in a collision it will, independently of all else, transmit at the end of each additional period with probability p—the so-called Aloha protocol (because it was first instituted at the University of Hawaii). We will show that such a system is asymptotically unstable in the sense that the number of successful transmissions will, with probability 1, be finite.

To begin let X_n denote the number of messages in the facility at the beginning of the nth period, and note that $\{X_n, n \geqslant 0\}$ is a Markov chain. Now for $k \geqslant 0$ define the indicator variables I_k by

$$I_k = \begin{cases} 1, & \text{if the first time that the chain departs state } k \text{ it} \\ & \text{directly goes to state } k-1 \\ 0, & \text{otherwise} \end{cases}$$

and let it be 0 if the system is never in state $k, k \geqslant 0$. (For instance, if the successive states are $0, 1, 3, 4, \ldots$, then $I_3 = 0$ since when the chain first departs state 3 it goes to state 4; whereas, if they are $0, 3, 3, 2, \ldots$, then $I_3 = 1$ since this time it goes to state 2.)

Now,

$$E\left[\sum_{k=0}^{\infty} I_k\right] = \sum_{k=0}^{\infty} E[I_k]$$
$$= \sum_{k=0}^{\infty} P\{I_k = 1\}$$
$$\leqslant \sum_{k=0}^{\infty} P\{I_k = 1 | k \text{ is ever visited}\} \tag{4.6}$$

Now, $P\{I_k = 1|k \text{ is ever visited}\}$ is the probability that when state k is departed the next state is $k-1$. That is, it is the conditional probability that a transition from k is to $k-1$ given that it is not back into k, and so

$$P\{I_k = 1|k \text{ is ever visited}\} = \frac{P_{k,k-1}}{1 - P_{k,k}}$$

Because

$$P_{k,k-1} = a_0 kp(1-p)^{k-1},$$
$$P_{k,k} = a_0[1 - kp(1-p)^{k-1}] + a_1(1-p)^k$$

which is seen by noting that if there are k messages present on the beginning of a day, then (a) there will be $k-1$ at the beginning of the next day if there are no new messages that day and exactly one of the k messages transmits; and (b) there will be k at the beginning of the next day if either

(i) there are no new messages and it is not the case that exactly one of the existing k messages transmits, or

(ii) there is exactly one new message (which automatically transmits) and none of the other k messages transmits.

Substitution of the preceding into Eq. (4.6) yields

$$E\left[\sum_{k=0}^{\infty} I_k\right] \leqslant \sum_{k=0}^{\infty} \frac{a_0 kp(1-p)^{k-1}}{1 - a_0[1 - kp(1-p)^{k-1}] - a_1(1-p)^k}$$
$$< \infty$$

where the convergence follows by noting that when k is large the denominator of the expression in the preceding sum converges to $1 - a_0$ and so the convergence or divergence of the sum is determined by whether or not the sum of the terms in the numerator converge and $\sum_{k=0}^{\infty} k(1-p)^{k-1} < \infty$.

Hence, $E[\sum_{k=0}^{\infty} I_k] < \infty$, which implies that $\sum_{k=0}^{\infty} I_k < \infty$ with probability 1 (for if there was a positive probability that $\sum_{k=0}^{\infty} I_k$ could be ∞, then its mean would be ∞). Hence, with probability 1, there will be only a finite number of states that are

initially departed via a successful transmission; or equivalently, there will be some finite integer N such that whenever there are N or more messages in the system, there will never again be a successful transmission. From this (and the fact that such higher states will eventually be reached—why?) it follows that, with probability 1, there will only be a finite number of successful transmissions. ■

Remark. For a (slightly less than rigorous) probabilistic proof of Stirling's approximation, let $X_1, X_2, \ldots$ be independent Poisson random variables each having mean 1. Let $S_n = \sum_{i=1}^{n} X_i$, and note that both the mean and variance of S_n are equal to n. Now,

$$\begin{aligned} P\{S_n = n\} &= P\{n-1 < S_n \leqslant n\} \\ &= P\{-1/\sqrt{n} < (S_n - n)/\sqrt{n} \leqslant 0\} \\ &\approx \int_{-1/\sqrt{n}}^{0} (2\pi)^{-1/2} e^{-x^2/2}\, dx \qquad \text{when } n \text{ is large, by the central limit theorem} \\ &\approx (2\pi)^{-1/2}(1/\sqrt{n}) \\ &= (2\pi n)^{-1/2} \end{aligned}$$

But S_n is Poisson with mean n, and so

$$P\{S_n = n\} = \frac{e^{-n} n^n}{n!}$$

Hence, for n large

$$\frac{e^{-n} n^n}{n!} \approx (2\pi n)^{-1/2}$$

or, equivalently

$$n! \approx n^{n+1/2} e^{-n} \sqrt{2\pi}$$

which is Stirling's approximation.

4.4 Long-Run Proportions and Limiting Probabilities

For pairs of states $i \neq j$, let $f_{i,j}$ denote the probability that the Markov chain, starting in state i, will ever make a transition into state j. That is,

$$f_{i,j} = P(X_n = j \text{ for some } n > 0 | X_0 = i)$$

We then have the following result.

Proposition 4.3. *If i is recurrent and i communicates with j, then $f_{i,j} = 1$.*

Proof. Because i and j communicate there is a value n such that $P_{i,j}^n > 0$. Let $X_0 = i$ and say that the first opportunity is a success if $X_n = j$, and note that the first opportunity is a success with probability $P_{i,j}^n > 0$. If the first opportunity is not a success then consider the next time (after time n) that the chain enters state i. (Because state i is recurrent we can be certain that it will eventually reenter state i after time n.) Say that the second opportunity is a success if n time periods later the Markov chain is in state j. If the second opportunity is not a success then wait until the next time the chain enters state i and say that the third opportunity is a success if n time periods later the Markov chain is in state j. Continuing in this manner, we can define an unlimited number of opportunities, each of which is a success with the same positive probability $P_{i,j}^n$. Because the number of opportunities until the first success occurs is geometric with parameter $P_{i,j}^n$, it follows that with probability 1 a success will eventually occur and so, with probability 1, state j will eventually be entered. ■

If state j is recurrent, let m_j denote the expected number of transitions that it takes the Markov chain when starting in state j to return to that state. That is, with

$$N_j = \min\{n > 0 : X_n = j\}$$

equal to the number of transitions until the Markov chain makes a transition into state j,

$$m_j = E[N_j | X_0 = j]$$

Definition. Say that the recurrent state j is *positive recurrent* if $m_j < \infty$ and say that it is *null recurrent* if $m_j = \infty$.

Now suppose that the Markov chain is irreducible and recurrent. In this case we now show that the long-run proportion of time that the chain spends in state j is equal to $1/m_j$. That is, letting π_j denote the long-run proportion of time that the Markov chain is in state j, we have the following proposition.

Proposition 4.4. *If the Markov chain is irreducible and recurrent, then for any initial state*

$$\pi_j = 1/m_j$$

Proof. Suppose that the Markov chain starts in state i, and let T_1 denote the number of transitions until the chain enters state j; then let T_2 denote the additional number of transitions from time T_1 until the Markov chain next enters state j; then let T_3 denote the additional number of transitions from time $T_1 + T_2$ until the Markov chain next enters state j, and so on. Note that T_1 is finite because Proposition 4.3 tells us that with probability 1 a transition into j will eventually occur. Also, for $n \geq 2$, because T_n is the number of transitions between the $(n-1)$th and the nth transition into state j, it follows from the Markovian property that $T_2, T_3, \ldots$ are independent and identically distributed with mean m_j. Because the nth transition into state j occurs at time $T_1 +$

$\ldots + T_n$ we obtain that π_j, the long-run proportion of time that the chain is in state j, is

$$\begin{aligned}\pi_j &= \lim_{n\to\infty} \frac{n}{\sum_{i=1}^n T_i} \\ &= \lim_{n\to\infty} \frac{1}{\frac{1}{n}\sum_{i=1}^n T_i} \\ &= \lim_{n\to\infty} \frac{1}{\frac{T_1}{n} + \frac{T_2+\ldots+T_n}{n}} \\ &= \frac{1}{m_j}\end{aligned}$$

where the last equality follows because $\lim_{n\to\infty} T_1/n = 0$ and, from the strong law of large numbers, $\lim_{n\to\infty} \frac{T_2+\ldots+T_n}{n} = \lim_{n\to\infty} \frac{T_2+\ldots+T_n}{n-1}\,\frac{n-1}{n} = m_j$. ■

Because $m_j < \infty$ is equivalent to $1/m_j > 0$, it follows from the preceding that state j is positive recurrent if and only if $\pi_j > 0$. We now exploit this to show that positive recurrence is a class property.

Proposition 4.5. *If i is positive recurrent and $i \leftrightarrow j$ then j is positive recurrent.*

Proof. Suppose that i is positive recurrent and that $i \leftrightarrow j$. Now, let n be such that $P_{i,j}^n > 0$. Because π_i is the long-run proportion of time that the chain is in state i, and $P_{i,j}^n$ is the long-run proportion of time when the Markov chain is in state i that it will be in state j after n transitions

$$\begin{aligned}\pi_i P_{i,j}^n &= \text{long-run proportion of time the chain is in } i \\ &\qquad \text{and will be in } j \text{ after } n \text{ transitions} \\ &= \text{long-run proportion of time the chain is in } j \\ &\qquad \text{and was in } i \text{ } n \text{ transitions ago} \\ &\le \text{long-run proportion of time the chain is in } j\end{aligned}$$

Hence, $\pi_j \ge \pi_i P_{i,j}^n > 0$, showing that j is positive recurrent. ■

Remarks. **(i)** It follows from the preceding result that null recurrence is also a class property. For suppose that i is null recurrent and $i \leftrightarrow j$. Because i is recurrent and $i \leftrightarrow j$ we can conclude that j is recurrent. But if j were positive recurrent then by the preceding proposition i would also be positive recurrent. Because i is not positive recurrent, neither is j.

(ii) An irreducible finite state Markov chain must be positive recurrent. For we know that such a chain must be recurrent; hence, all its states are either positive recurrent or null recurrent. If they were null recurrent then all the long run proportions would equal 0, which is impossible when there are only a finite number of states. Consequently, we can conclude that the chain is positive recurrent.

(iii) The classical example of a null recurrent Markov chain is the one dimensional symmetric random walk of Example 4.18. One way to show it is null recurrent is to argue that the mean time to return to a state is infinite. (For another argument, see Exercise 39.) To show this, let $m_{i,j}$ denote the mean number of transitions, starting in state i, until a transition into state j occurs. Now, in Example 3.16, it was shown that the mean number of transitions to go from state 1 to either 0 or n is $n-1$, implying that

$$m_{1,0} \geq n-1.$$

As the preceding is true for all n, we obtain upon letting $n \to \infty$ that

$$m_{1,0} = \infty.$$

Conditioning on the result of the first bet gives

$$m_{0,0} = 1 + m_{1,0}\frac{1}{2} + m_{-1,0}\frac{1}{2}.$$

Hence, $m_{0,0} = \infty$, establishing that the symmetric random walk is null recurrent. ■

To determine the long-run proportions $\{\pi_j, j \geq 1\}$, note, because π_i is the long-run proportion of transitions that come from state i, that

$$\pi_i P_{i,j} = \text{long-run proportion of transitions that go from state } i \text{ to state } j$$

Summing the preceding over all i now yields that

$$\pi_j = \sum_i \pi_i P_{i,j}$$

Indeed, the following important theorem can be proven.

Theorem 4.1. *Consider an irreducible Markov chain. If the chain is positive recurrent then the long-run proportions are the unique solution of the equations*

$$\begin{aligned} \pi_j &= \sum_i \pi_i P_{i,j}, \quad j \geq 1 \\ \sum_j \pi_j &= 1 \end{aligned} \tag{4.7}$$

Moreover, if there is no solution of the preceding linear equations, then the Markov chain is either transient or null recurrent and all $\pi_j = 0$.

Example 4.22. Consider Example 4.1, in which we assume that if it rains today, then it will rain tomorrow with probability α; and if it does not rain today, then it will rain

tomorrow with probability β. If we say that the state is 0 when it rains and 1 when it does not rain, then by Theorem 4.1 the long-run proportions π_0 and π_1 are given by

$$\begin{aligned}\pi_0 &= \alpha\pi_0 + \beta\pi_1,\\ \pi_1 &= (1-\alpha)\pi_0 + (1-\beta)\pi_1,\\ \pi_0 + \pi_1 &= 1\end{aligned}$$

which yields that

$$\pi_0 = \frac{\beta}{1+\beta-\alpha}, \quad \pi_1 = \frac{1-\alpha}{1+\beta-\alpha}$$

For example if $\alpha = 0.7$ and $\beta = 0.4$, then the long-run proportion of rain is $\pi_0 = \frac{4}{7} = 0.571$. ■

Example 4.23. Consider Example 4.3 in which the mood of an individual is considered as a three-state Markov chain having a transition probability matrix

$$\mathbf{P} = \begin{Vmatrix} 0.5 & 0.4 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{Vmatrix}$$

In the long run, what proportion of time is the process in each of the three states?

Solution: The long run proportions $\pi_i, i = 0, 1, 2$, are obtained by solving the set of equations in Eq. (4.7). In this case these equations are

$$\begin{aligned}\pi_0 &= 0.5\pi_0 + 0.3\pi_1 + 0.2\pi_2,\\ \pi_1 &= 0.4\pi_0 + 0.4\pi_1 + 0.3\pi_2,\\ \pi_2 &= 0.1\pi_0 + 0.3\pi_1 + 0.5\pi_2,\\ \pi_0 + \pi_1 + \pi_2 &= 1\end{aligned}$$

Solving yields

$$\pi_0 = \tfrac{21}{62}, \quad \pi_1 = \tfrac{23}{62}, \quad \pi_2 = \tfrac{18}{62}$$ ■

Example 4.24 (A Model of Class Mobility)**.** A problem of interest to sociologists is to determine the proportion of society that has an upper- or lower-class occupation. One possible mathematical model would be to assume that transitions between social classes of the successive generations in a family can be regarded as transitions of a Markov chain. That is, we assume that the occupation of a child depends only on his or her parent's occupation. Let us suppose that such a model is appropriate and that the transition probability matrix is given by

$$\mathbf{P} = \begin{Vmatrix} 0.45 & 0.48 & 0.07 \\ 0.05 & 0.70 & 0.25 \\ 0.01 & 0.50 & 0.49 \end{Vmatrix} \tag{4.8}$$

That is, for instance, we suppose that the child of a middle-class worker will attain an upper-, middle-, or lower-class occupation with respective probabilities 0.05, 0.70, 0.25.

The long-run proportions π_i thus satisfy

$$\begin{aligned}\pi_0 &= 0.45\pi_0 + 0.05\pi_1 + 0.01\pi_2,\\ \pi_1 &= 0.48\pi_0 + 0.70\pi_1 + 0.50\pi_2,\\ \pi_2 &= 0.07\pi_0 + 0.25\pi_1 + 0.49\pi_2,\\ \pi_0 + \pi_1 + \pi_2 &= 1\end{aligned}$$

Hence,

$$\pi_0 = 0.07, \quad \pi_1 = 0.62, \quad \pi_2 = 0.31$$

In other words, a society in which social mobility between classes can be described by a Markov chain with transition probability matrix given by Eq. (4.8) has, in the long run, 7 percent of its people in upper-class jobs, 62 percent of its people in middle-class jobs, and 31 percent in lower-class jobs.

Example 4.25 (The Hardy–Weinberg Law and a Markov Chain in Genetics)**.** Consider a large population of individuals, each of whom possesses a particular pair of genes, of which each individual gene is classified as being of type A or type a. Assume that the proportions of individuals whose gene pairs are AA, aa, or Aa are, respectively, p_0, q_0, and r_0 ($p_0 + q_0 + r_0 = 1$). When two individuals mate, each contributes one of his or her genes, chosen at random, to the resultant offspring. Assuming that the mating occurs at random, in that each individual is equally likely to mate with any other individual, we are interested in determining the proportions of individuals in the next generation whose genes are AA, aa, or Aa. Calling these proportions p, q, and r, they are easily obtained by focusing attention on an individual of the next generation and then determining the probabilities for the gene pair of that individual.

To begin, note that randomly choosing a parent and then randomly choosing one of its genes is equivalent to just randomly choosing a gene from the total gene population. By conditioning on the gene pair of the parent, we see that a randomly chosen gene will be type A with probability

$$\begin{aligned}P\{A\} &= P\{A|AA\}p_0 + P\{A|aa\}q_0 + P\{A|Aa\}r_0\\ &= p_0 + r_0/2\end{aligned}$$

Similarly, it will be type a with probability

$$P\{a\} = q_0 + r_0/2$$

Thus, under random mating a randomly chosen member of the next generation will be type AA with probability p, where

$$p = P\{A\}P\{A\} = (p_0 + r_0/2)^2$$

Similarly, the randomly chosen member will be type aa with probability

$$q = P\{a\}P\{a\} = (q_0 + r_0/2)^2$$

and will be type Aa with probability

$$r = 2P\{A\}P\{a\} = 2(p_0 + r_0/2)(q_0 + r_0/2)$$

Since each member of the next generation will independently be of each of the three gene types with probabilities p, q, r, it follows that the percentages of the members of the next generation that are of type AA, aa, or Aa are respectively p, q, and r.

If we now consider the total gene pool of this next generation, then $p + r/2$, the fraction of its genes that are A, will be unchanged from the previous generation. This follows either by arguing that the total gene pool has not changed from generation to generation or by the following simple algebra:

$$\begin{aligned} p + r/2 &= (p_0 + r_0/2)^2 + (p_0 + r_0/2)(q_0 + r_0/2) \\ &= (p_0 + r_0/2)[p_0 + r_0/2 + q_0 + r_0/2] \\ &= p_0 + r_0/2 \quad \text{since } p_0 + r_0 + q_0 = 1 \\ &= P\{A\} \end{aligned} \tag{4.9}$$

Thus, the fractions of the gene pool that are A and a are the same as in the initial generation. From this it follows that, under random mating, in all successive generations after the initial one the percentages of the population having gene pairs AA, aa, and Aa will remain fixed at the values p, q, and r. This is known as the *Hardy–Weinberg law*.

Suppose now that the gene pair population has stabilized in the percentages p, q, r, and let us follow the genetic history of a single individual and her descendants. (For simplicity, assume that each individual has exactly one offspring.) So, for a given individual, let X_n denote the genetic state of her descendant in the nth generation. The transition probability matrix of this Markov chain, namely,

$$\begin{array}{c} \\ AA \\ aa \\ Aa \end{array} \begin{array}{c} \begin{array}{ccc} AA & aa & Aa \end{array} \\ \left\| \begin{array}{ccc} p + \frac{r}{2} & 0 & q + \frac{r}{2} \\ 0 & q + \frac{r}{2} & p + \frac{r}{2} \\ \frac{p}{2} + \frac{r}{4} & \frac{q}{2} + \frac{r}{4} & \frac{p}{2} + \frac{q}{2} + \frac{r}{2} \end{array} \right\| \end{array}$$

is easily verified by conditioning on the state of the randomly chosen mate. It is quite intuitive (why?) that the limiting probabilities for this Markov chain (which also equal the fractions of the individual's descendants that are in each of the three genetic states) should just be p, q, and r. To verify this we must show that they satisfy Theorem 4.1.

Because one of the equations in Theorem 4.1 is redundant, it suffices to show that

$$p = p\left(p + \frac{r}{2}\right) + r\left(\frac{p}{2} + \frac{r}{4}\right) = \left(p + \frac{r}{2}\right)^2,$$
$$q = q\left(q + \frac{r}{2}\right) + r\left(\frac{q}{2} + \frac{r}{4}\right) = \left(q + \frac{r}{2}\right)^2,$$
$$p + q + r = 1$$

But this follows from Eq. (4.9), and thus the result is established. ■

Example 4.26. Suppose that a production process changes states in accordance with an irreducible, positive recurrent Markov chain having transition probabilities $P_{ij}, i, j = 1, \ldots, n$, and suppose that certain of the states are considered acceptable and the remaining unacceptable. Let A denote the acceptable states and A^c the unacceptable ones. If the production process is said to be "up" when in an acceptable state and "down" when in an unacceptable state, determine

1. the rate at which the production process goes from up to down (that is, the rate of breakdowns);
2. the average length of time the process remains down when it goes down; and
3. the average length of time the process remains up when it goes up.

Solution: Let $\pi_k, k = 1, \ldots, n$, denote the long-run proportions. Now for $i \in A$ and $j \in A^c$ the rate at which the process enters state j from state i is

$$\text{rate enter } j \text{ from } i = \pi_i P_{ij}$$

and so the rate at which the production process enters state j from an acceptable state is

$$\text{rate enter } j \text{ from } A = \sum_{i \in A} \pi_i P_{ij}$$

Hence, the rate at which it enters an unacceptable state from an acceptable one (which is the rate at which breakdowns occur) is

$$\text{rate breakdowns occur} = \sum_{j \in A^c} \sum_{i \in A} \pi_i P_{ij} \tag{4.10}$$

Now let $\bar{U}$ and $\bar{D}$ denote the average time the process remains up when it goes up and down when it goes down. Because there is a single breakdown every $\bar{U} + \bar{D}$ time units on the average, it follows heuristically that

$$\text{rate at which breakdowns occur} = \frac{1}{\bar{U} + \bar{D}}$$

and so from Eq. (4.10),

$$\frac{1}{\bar{U} + \bar{D}} = \sum_{j \in A^c} \sum_{i \in A} \pi_i P_{ij} \tag{4.11}$$

To obtain a second equation relating $\bar{U}$ and $\bar{D}$, consider the percentage of time the process is up, which, of course, is equal to $\sum_{i\in A}\pi_i$. However, since the process is up on the average $\bar{U}$ out of every $\bar{U}+\bar{D}$ time units, it follows (again somewhat heuristically) that the

$$\text{proportion of up time} = \frac{\bar{U}}{\bar{U}+\bar{D}}$$

and so

$$\frac{\bar{U}}{\bar{U}+\bar{D}} = \sum_{i\in A}\pi_i \tag{4.12}$$

Hence, from Eqs. (4.11) and (4.12) we obtain

$$\begin{aligned}
\bar{U} &= \frac{\sum_{i\in A}\pi_i}{\sum_{j\in A^c}\sum_{i\in A}\pi_i P_{ij}},\\
\bar{D} &= \frac{1-\sum_{i\in A}\pi_i}{\sum_{j\in A^c}\sum_{i\in A}\pi_i P_{ij}}\\
&= \frac{\sum_{i\in A^c}\pi_i}{\sum_{j\in A^c}\sum_{i\in A}\pi_i P_{ij}}
\end{aligned}$$

For example, suppose the transition probability matrix is

$$\mathbf{P} = \begin{Vmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{2} & 0 \\ 0 & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{2} \end{Vmatrix}$$

where the acceptable (up) states are 1, 2 and the unacceptable (down) ones are 3, 4. The long-run proportions satisfy

$$\begin{aligned}
\pi_1 &= \pi_1\tfrac{1}{4} + \pi_3\tfrac{1}{4} + \pi_4\tfrac{1}{4},\\
\pi_2 &= \pi_1\tfrac{1}{4} + \pi_2\tfrac{1}{4} + \pi_3\tfrac{1}{4} + \pi_4\tfrac{1}{4},\\
\pi_3 &= \pi_1\tfrac{1}{2} + \pi_2\tfrac{1}{2} + \pi_3\tfrac{1}{4},\\
\pi_1 + \pi_2 + \pi_3 + \pi_4 &= 1
\end{aligned}$$

These solve to yield

$$\pi_1 = \tfrac{3}{16}, \quad \pi_2 = \tfrac{1}{4}, \quad \pi_3 = \tfrac{14}{48}, \quad \pi_4 = \tfrac{13}{48}$$

and thus

$$\begin{aligned}\text{rate of breakdowns} &= \pi_1(P_{13} + P_{14}) + \pi_2(P_{23} + P_{24}) \\ &= \tfrac{9}{32}, \\ \bar{U} = \tfrac{14}{9} \quad &\text{and} \quad \bar{D} = 2\end{aligned}$$

Hence, on the average, breakdowns occur about $\frac{9}{32}$ (or 28 percent) of the time. They last, on the average, 2 time units, and then there follows a stretch of (on the average) $\frac{14}{9}$ time units when the system is up. ■

The long run proportions π_j, $j \geqslant 0$, are often called *stationary* probabilities. The reason being that if the initial state is chosen according to the probabilities π_j, $j \geqslant 0$, then the probability of being in state j at any time n is also equal to π_j. That is, if

$$P\{X_0 = j\} = \pi_j, \quad j \geqslant 0$$

then

$$P\{X_n = j\} = \pi_j \quad \text{for all } n,\ j \geqslant 0$$

The preceding is easily proven by induction, for it is true when $n = 0$ and if we suppose it true for $n - 1$, then writing

$$\begin{aligned}P\{X_n = j\} &= \sum_i P\{X_n = j | X_{n-1} = i\} P\{X_{n-1} = i\} \\ &= \sum_i P_{ij}\pi_i \quad \text{by the induction hypothesis} \\ &= \pi_j \qquad\qquad \text{by Theorem 4.1}\end{aligned}$$

Example 4.27. Suppose the numbers of families that check into a hotel on successive days are independent Poisson random variables with mean λ. Also suppose that the number of days that a family stays in the hotel is a geometric random variable with parameter $p, 0 < p < 1$. (Thus, a family who spent the previous night in the hotel will, independently of how long they have already spent in the hotel, check out the next day with probability p.) Also suppose that all families act independently of each other. Under these conditions it is easy to see that if X_n denotes the number of families that are checked in the hotel at the beginning of day n then $\{X_n, n \geq 0\}$ is a Markov chain. Find

(a) the transition probabilities of this Markov chain;
(b) $E[X_n | X_0 = i]$;
(c) the stationary probabilities of this Markov chain.

Solution: (a) To find $P_{i,j}$, suppose there are i families checked into the hotel at the beginning of a day. Because each of these i families will stay for another day with probability $q = 1 - p$ it follows that R_i, the number of these families that remain another day, is a binomial (i, q) random variable. So, letting N be the number of new families that check in that day, we see that

$$P_{i,j} = P(R_i + N = j)$$

Conditioning on R_i and using that N is Poisson with mean λ, we obtain

$$\begin{aligned}
P_{i,j} &= \sum_{k=0}^{i} P(R_i + N = j | R_i = k) \binom{i}{k} q^k p^{i-k} \\
&= \sum_{k=0}^{i} P(N = j - k | R_i = k) \binom{i}{k} q^k p^{i-k} \\
&= \sum_{k=0}^{\min(i,j)} P(N = j - k) \binom{i}{k} q^k p^{i-k} \\
&= \sum_{k=0}^{\min(i,j)} e^{-\lambda} \frac{\lambda^{j-k}}{(j-k)!} \binom{i}{k} q^k p^{i-k}
\end{aligned}$$

(b) Using the preceding representation $R_i + N$ for the next state from state i, we see that

$$E[X_n | X_{n-1} = i] = E[R_i + N] = iq + \lambda$$

Consequently,

$$E[X_n | X_{n-1}] = X_{n-1} q + \lambda$$

Taking expectations of both sides yields

$$E[X_n] = \lambda + q E[X_{n-1}]$$

Iterating the preceding gives

$$\begin{aligned}
E[X_n] &= \lambda + q E[X_{n-1}] \\
&= \lambda + q(\lambda + q E[X_{n-2}]) \\
&= \lambda + q\lambda + q^2 E[X_{n-2}] \\
&= \lambda + q\lambda + q^2(\lambda + q E[X_{n-3}]) \\
&= \lambda + q\lambda + q^2\lambda + q^3 E[X_{n-3}]
\end{aligned}$$

showing that

$$E[X_n] = \lambda \left(1 + q + q^2 + \ldots + q^{n-1}\right) + q^n E[X_0]$$

and yielding the result

$$E[X_n | X_0 = i] = \frac{\lambda(1 - q^n)}{p} + q^n i$$

(c) To find the stationary probabilities we will not directly use the complicated transition probabilities derived in part (a). Rather we will make use of the fact that

the stationary probability distribution is the only distribution on the initial state that results in the next state having the same distribution. Now, suppose that the initial state X_0 has a Poisson distribution with mean α. That is, assume that the number of families initially in the hotel is Poisson with mean α. Let R denote the number of these families that remain in the hotel at the beginning of the next day. Then, using the result of Example 3.24 that if each of a Poisson distributed (with mean α) number of events occurs with probability q, then the total number of these events that occur is Poisson distributed with mean αq, it follows that R is a Poisson random variable with mean αq. In addition, the number of new families that check in during the day, call it N, is Poisson with mean λ, and is independent of R. Hence, since the sum of independent Poisson random variables is also Poisson distributed, it follows that $R + N$, the number of guests at the beginning of the next day, is Poisson with mean $\lambda + \alpha q$. Consequently, if we choose α so that

$$\alpha = \lambda + \alpha q$$

then the distribution of X_1 would be the same as that of X_0. But this means that when the initial distribution of X_0 is Poisson with mean $\alpha = \frac{\lambda}{p}$, then so is the distribution of X_1, implying that this is the stationary distribution. That is, the stationary probabilities are

$$\pi_i = e^{-\lambda/p}(\lambda/p)^i/i!, \quad i \geq 0$$

The preceding model has an important generalization. Namely, consider an organization whose workers are of r distinct types. For instance, the organization could be a law firm and its lawyers could either be juniors, associates, or partners. Suppose that a worker who is currently type i will in the next period become type j with probability $q_{i,j}$ for $j = 1, \ldots, r$ or will leave the organization with probability $1 - \sum_{j=1}^{r} q_{i,j}$. In addition, suppose that new workers are hired each period, and that the numbers of types $1, \ldots, r$ workers hired are independent Poisson random variables with means $\lambda_1, \ldots, \lambda_r$. If we let $\mathbf{X}_n = (X_n(1), \ldots, X_n(r))$, where $X_n(i)$ is the number of type i workers in the organization at the beginning of period n, then $\mathbf{X}_n, n \geq 0$ is a Markov chain. To compute its stationary probability distribution, suppose that the initial state is chosen so that the number of workers of different types are independent Poisson random variables, with α_i being the mean number of type i workers. That is, suppose that $X_0(1), \ldots, X_0(r)$ are independent Poisson random variables with respective means $\alpha_1, \ldots, \alpha_r$. Also, let $N_j, j = 1, \ldots, r$, be the number of new type j workers hired during the initial period. Now, fix i, and for $j = 1, \ldots, r$, let $M_i(j)$ be the number of the $X_0(i)$ type i workers who become type j in the next period. Then because each of the Poisson number $X_0(i)$ of type i workers will independently become type j with probability $q_{i,j}, j = 1, \ldots, r$, it follows from the remarks following Example 3.24 that $M_i(1), \ldots, M_i(r)$ are independent Poisson random variables with $M_i(j)$ having mean $a_i q_{i,j}$. Because $X_0(1), \ldots, X_0(r)$ are, by assumption, independent, we can also conclude that the random variables $M_i(j), i, j = 1, \ldots, r$ are all independent. Because the sum of independent Poisson random variables is also Poisson

distributed, the preceding yields that the random variables

$$X_1(j) = N_j + \sum_{i=1}^{r} M_i(j), \quad j = 1, \ldots, r$$

are independent Poisson random variables with means

$$E[X_1(j)] = \lambda_j + \sum_{i=1}^{r} \alpha_i q_{i,j}$$

Hence, if $\alpha_1, \ldots, \alpha_r$ satisfied

$$\alpha_j = \lambda_j + \sum_{i=1}^{r} \alpha_i q_{i,j}, \quad j = 1, \ldots, r$$

then $\mathbf{X}_1$ would have the same distribution as $\mathbf{X}_0$. Consequently, if we let $\alpha_1^o, \ldots, \alpha_r^o$ be such that

$$\alpha_j^o = \lambda_j + \sum_{i=1}^{r} \alpha_i^o q_{i,j}, \quad j = 1, \ldots, r$$

then the stationary distribution of the Markov chain is the distribution that takes the number of workers in each type to be independent Poisson random variables with means $\alpha_1^o, \ldots, \alpha_r^o$. That is, the long run proportions are

$$\pi_{k_1,\ldots,k_r} = \prod_{i=1}^{r} e^{-\alpha_i^o} (\alpha_i^o)^{k_i} / k_i!$$

It can be shown that there will be such values α_j^o, $j = 1, \ldots, r$, provided that, with probability 1, each worker eventually leaves the organization. Also, because there is a unique stationary distribution, there can only be one such set of values. ■

The following example exploits the relationship $m_i = 1/\pi_i$, which states that the mean time between visits to a state is the inverse of the long run proportion of time the chain is in that state, to obtain a method for computing the mean time until a specified pattern appears when the data constitutes the successive states of a Markov chain.

Example 4.28 (Mean Pattern Times in Markov Chain Generated Data)**.** Consider an irreducible Markov chain $\{X_n, n \geqslant 0\}$ with transition probabilities $P_{i,j}$ and stationary probabilities π_j, $j \geqslant 0$. Starting in state r, we are interested in determining the expected number of transitions until the pattern $i_1, i_2, \ldots, i_k$ appears. That is, with

$$N(i_1, i_2, \ldots, i_k) = \min\{n \geqslant k\colon X_{n-k+1} = i_1, \ldots, X_n = i_k\}$$

we are interested in

$$E[N(i_1, i_2, \ldots, i_k) | X_0 = r]$$

Note that even if $i_1 = r$, the initial state X_0 is not considered part of the pattern sequence.

Let $\mu(i, i_1)$ be the mean number of transitions for the chain to enter state i_1, given that the initial state is i, $i \geqslant 0$. The quantities $\mu(i, i_1)$ can be determined as the solution of the following set of equations, obtained by conditioning on the first transition out of state i:

$$\mu(i, i_1) = 1 + \sum_{j \neq i_1} P_{i,j}\mu(j, i_1), \quad i \geqslant 0$$

For the Markov chain $\{X_n, n \geqslant 0\}$ associate a corresponding Markov chain, which we will refer to as the k-chain, whose state at any time is the sequence of the most recent k states of the original chain. (For instance, if $k = 3$ and $X_2 = 4, X_3 = 1, X_4 = 1$, then the state of the k-chain at time 4 is (4, 1, 1).) Let $\pi(j_1, \ldots, j_k)$ be the stationary probabilities for the k-chain. Because $\pi(j_1, \ldots, j_k)$ is the proportion of time that the state of the original Markov chain k units ago was j_1 and the following $k - 1$ states, in sequence, were $j_2, \ldots, j_k$, we can conclude that

$$\pi(j_1, \ldots, j_k) = \pi_{j_1} P_{j_1,j_2} \cdots P_{j_{k-1},j_k}$$

Moreover, because the mean number of transitions between successive visits of the k-chain to the state $i_1, i_2, \ldots, i_k$ is equal to the inverse of the stationary probability of that state, we have that

$$E[\text{number of transitions between visits to } i_1, i_2, \ldots, i_k] = \frac{1}{\pi(i_1, \ldots, i_k)} \tag{4.13}$$

Let $A(i_1, \ldots, i_m)$ be the additional number of transitions needed until the pattern appears, given that the first m transitions have taken the chain into states $X_1 = i_1, \ldots, X_m = i_m$.

We will now consider whether the pattern has overlaps, where we say that the pattern $i_1, i_2, \ldots, i_k$ has an overlap of size $j, j < k$, if the sequence of its final j elements is the same as that of its first j elements. That is, it has an overlap of size j if

$$(i_{k-j+1}, \ldots, i_k) = (i_1, \ldots, i_j), \quad j < k$$

Case 1. The pattern $i_1, i_2, \ldots, i_k$ has no overlaps.
Because there is no overlap, Eq. (4.13) yields

$$E[N(i_1, i_2, \ldots, i_k) | X_0 = i_k] = \frac{1}{\pi(i_1, \ldots, i_k)}$$

Because the time until the pattern occurs is equal to the time until the chain enters state i_1 plus the additional time, we may write

$$E[N(i_1, i_2, \ldots, i_k) | X_0 = i_k] = \mu(i_k, i_1) + E[A(i_1)]$$

The preceding two equations imply

$$E[A(i_1)] = \frac{1}{\pi(i_1, \ldots, i_k)} - \mu(i_k, i_1)$$

Using that

$$E[N(i_1, i_2, \ldots, i_k)|X_0 = r] = \mu(r, i_1) + E[A(i_1)]$$

gives the result

$$E[N(i_1, i_2, \ldots, i_k)|X_0 = r] = \mu(r, i_1) + \frac{1}{\pi(i_1, \ldots, i_k)} - \mu(i_k, i_1)$$

where

$$\pi(i_1, \ldots, i_k) = \pi_{i_1} P_{i_1,i_2} \cdots P_{i_{k-1},i_k}$$

Case 2. Now suppose that the pattern has overlaps and let its largest overlap be of size s. In this case the number of transitions between successive visits of the k-chain to the state $i_1, i_2, \ldots, i_k$ is equal to the additional number of transitions of the original chain until the pattern appears given that it has already made s transitions with the results $X_1 = i_1, \ldots, X_s = i_s$. Therefore, from Eq. (4.13)

$$E[A(i_1, \ldots, i_s)] = \frac{1}{\pi(i_1, \ldots, i_k)}$$

But because

$$N(i_1, i_2, \ldots, i_k) = N(i_1, \ldots, i_s) + A(i_1, \ldots, i_s)$$

we have

$$E[N(i_1, i_2, \ldots, i_k)|X_0 = r] = E[N(i_1, i_2, \ldots, i_s)|X_0 = r] + \frac{1}{\pi(i_1, \ldots, i_k)}$$

We can now repeat the same procedure on the pattern $i_1, \ldots, i_s$, continuing to do so until we reach one that has no overlap, and then apply the result from Case 1.

For instance, suppose the desired pattern is 1, 2, 3, 1, 2, 3, 1, 2. Then

$$E[N(1, 2, 3, 1, 2, 3, 1, 2)|X_0 = r] = E[N(1, 2, 3, 1, 2)|X_0 = r] + \frac{1}{\pi(1, 2, 3, 1, 2, 3, 1, 2)}$$

Because the largest overlap of the pattern (1, 2, 3, 1, 2) is of size 2, the same argument as in the preceding gives

$$E[N(1, 2, 3, 1, 2)|X_0 = r] = E[N(1, 2)|X_0 = r] + \frac{1}{\pi(1, 2, 3, 1, 2)}$$

Because the pattern (1, 2) has no overlap, we obtain from Case 1 that

$$E[N(1,2)|X_0 = r] = \mu(r,1) + \frac{1}{\pi(1,2)} - \mu(2,1)$$

Putting it together yields

$$E[N(1,2,3,1,2,3,1,2)|X_0 = r] = \mu(r,1) + \frac{1}{\pi_1 P_{1,2}} - \mu(2,1) + \frac{1}{\pi_1 P_{1,2}^2 P_{2,3} P_{3,1}} + \frac{1}{\pi_1 P_{1,2}^3 P_{2,3}^2 P_{3,1}^2}$$

If the generated data is a sequence of independent and identically distributed random variables, with each value equal to j with probability P_j, then the Markov chain has $P_{i,j} = P_j$. In this case, $\pi_j = P_j$. Also, because the time to go from state i to state j is a geometric random variable with parameter P_j, we have $\mu(i, j) = 1/P_j$. Thus, the expected number of data values that need be generated before the pattern 1, 2, 3, 1, 2, 3, 1, 2 appears would be

$$\frac{1}{P_1} + \frac{1}{P_1 P_2} - \frac{1}{P_1} + \frac{1}{P_1^2 P_2^2 P_3} + \frac{1}{P_1^3 P_2^3 P_3^2} = \frac{1}{P_1 P_2} + \frac{1}{P_1^2 P_2^2 P_3} + \frac{1}{P_1^3 P_2^3 P_3^2}$$ ■

The following result is quite useful.

Proposition 4.6. *Let $\{X_n, n \geqslant 1\}$ be an irreducible Markov chain with stationary probabilities π_j, $j \geqslant 0$, and let r be a bounded function on the state space. Then, with probability 1,*

$$\lim_{N\to\infty} \frac{\sum_{n=1}^{N} r(X_n)}{N} = \sum_{j=0}^{\infty} r(j)\pi_j$$

Proof. If we let $a_j(N)$ be the amount of time the Markov chain spends in state j during time periods $1, \ldots, N$, then

$$\sum_{n=1}^{N} r(X_n) = \sum_{j=0}^{\infty} a_j(N) r(j)$$

Since $a_j(N)/N \to \pi_j$ the result follows from the preceding upon dividing by N and then letting $N \to \infty$. ■

If we suppose that we earn a reward $r(j)$ whenever the chain is in state j, then Proposition 4.6 states that our average reward per unit time is $\sum_j r(j)\pi_j$.

Example 4.29. For the four state Bonus Malus automobile insurance system specified in Example 4.7, find the average annual premium paid by a policyholder whose yearly number of claims is a Poisson random variable with mean 1/2.

Solution: With $a_k = e^{-1/2}\frac{(1/2)^k}{k!}$, we have

$$a_0 = 0.6065, \quad a_1 = 0.3033, \quad a_2 = 0.0758$$

Therefore, the Markov chain of successive states has the following transition probability matrix:

$$\begin{Vmatrix} 0.6065 & 0.3033 & 0.0758 & 0.0144 \\ 0.6065 & 0.0000 & 0.3033 & 0.0902 \\ 0.0000 & 0.6065 & 0.0000 & 0.3935 \\ 0.0000 & 0.0000 & 0.6065 & 0.3935 \end{Vmatrix}$$

The stationary probabilities are given as the solution of

$$\begin{aligned} \pi_1 &= 0.6065\pi_1 + 0.6065\pi_2, \\ \pi_2 &= 0.3033\pi_1 + 0.6065\pi_3, \\ \pi_3 &= 0.0758\pi_1 + 0.3033\pi_2 + 0.6065\pi_4, \\ \pi_1 + \pi_2 + \pi_3 + \pi_4 &= 1 \end{aligned}$$

Rewriting the first three of these equations gives

$$\begin{aligned} \pi_2 &= \frac{1 - 0.6065}{0.6065}\pi_1, \\ \pi_3 &= \frac{\pi_2 - 0.3033\pi_1}{0.6065}, \\ \pi_4 &= \frac{\pi_3 - 0.0758\pi_1 - 0.3033\pi_2}{0.6065} \end{aligned}$$

or

$$\begin{aligned} \pi_2 &= 0.6488\pi_1, \\ \pi_3 &= 0.5697\pi_1, \\ \pi_4 &= 0.4900\pi_1 \end{aligned}$$

Using that $\sum_{i=1}^{4} \pi_i = 1$ gives the solution (rounded to four decimal places)

$$\pi_1 = 0.3692, \quad \pi_2 = 0.2395, \quad \pi_3 = 0.2103, \quad \pi_4 = 0.1809$$

Therefore, the average annual premium paid is

$$200\pi_1 + 250\pi_2 + 400\pi_3 + 600\pi_4 = 326.375$$ ■

4.4.1 Limiting Probabilities

In Example 4.8 we considered a two-state Markov chain with transition probability matrix

$$\mathbf{P} = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix}$$

and showed that

$$\mathbf{P}^{(4)} = \begin{pmatrix} 0.5749 & 0.4251 \\ 0.5668 & 0.4332 \end{pmatrix}$$

From this it follows that $\mathbf{P}^{(8)} = \mathbf{P}^{(4)} \cdot \mathbf{P}^{(4)}$ is given (to three significant places) by

$$\mathbf{P}^{(8)} = \begin{pmatrix} 0.571 & 0.429 \\ 0.571 & 0.429 \end{pmatrix}$$

Note that the matrix $\mathbf{P}^{(8)}$is almost identical to the matrix $\mathbf{P}^{(4)}$, and that each of the rows of $\mathbf{P}^{(8)}$ has almost identical values. Indeed, it seems that P_{ij}^n is converging to some value as $n \to \infty$, with this value not depending on i. Moreover, in Example 4.22 we showed that the long-run proportions for this chain are $\pi_0 = 4/7 \approx .571, \pi_1 = 3/7 \approx .429$, thus making it appear that these long-run proportions may also be limiting probabilities. Although this is indeed the case for the preceding chain, it is not always true that the long-run proportions are also limiting probabilities. To see why not, consider a two-state Markov chain having

$$P_{0,1} = P_{1,0} = 1$$

Because this Markov chain continually alternates between states 0 and 1, the long-run proportions of time it spends in these states are

$$\pi_0 = \pi_1 = 1/2$$

However,

$$P_{0,0}^n = \begin{cases} 1, & \text{if } n \text{ is even} \\ 0, & \text{if } n \text{ is odd} \end{cases}$$

and so $P_{0,0}^n$ does not have a limiting value as n goes to infinity. In general, a chain that can only return to a state in a multiple of $d > 1$ steps (where $d = 2$ in the preceding example) is said to be *periodic* and does not have limiting probabilities. However, for an irreducible chain that is not periodic, and such chains are called *aperiodic*, the limiting probabilities will always exist and will not depend on the initial state. Moreover, the limiting probability that the chain will be in state j will equal π_j, the long-run proportion of time the chain is in state j. That the limiting probabilities, when they exist, will equal the long-run proportions can be seen by letting

$$\alpha_j = \lim_{n \to \infty} P(X_n = j)$$

we now use the recursive identity (4.37) to determine $V_2(j)$ for each j; then $V_3(j)$ for each j; and so on, up to $V_n(j)$ for each j.

To obtain the maximizing sequence of states, we work in the reverse direction. Let j_n be the value (or any of the values if there are more than one) of j that maximizes $V_n(j)$. Thus j_n is the final state of a maximizing state sequence. Also, for $k < n$, let $i_k(j)$ be a value of i that maximizes $P_{i,j}V_k(i)$. Then

$$\begin{aligned}
&\max_{i_1,\ldots,i_n} P\{\mathbf{X}_n = (i_1,\ldots,i_n), \mathbf{S}^n = \mathbf{s}_n\} \\
&\quad = \max_j V_n(j) \\
&\quad = V_n(j_n) \\
&\quad = \max_{i_1,\ldots,i_{n-1}} P\{\mathbf{X}_n = (i_1,\ldots,i_{n-1}, j_n), \mathbf{S}^n = \mathbf{s}_n\} \\
&\quad = p(s_n|j_n)\max_i P_{i,j_n}V_{n-1}(i) \\
&\quad = p(s_n|j_n)P_{i_{n-1}(j_n),j_n}V_{n-1}(i_{n-1}(j_n))
\end{aligned}$$

Thus, $i_{n-1}(j_n)$ is the next to last state of the maximizing sequence. Continuing in this manner, the second from the last state of the maximizing sequence is $i_{n-2}(i_{n-1}(j_n))$, and so on.

The preceding approach to finding the most likely sequence of states given a prescribed sequence of signals is known as the *Viterbi Algorithm*.

Exercises

***1.** Three white and three black balls are distributed in two urns in such a way that each contains three balls. We say that the system is in state $i, i = 0, 1, 2, 3$, if the first urn contains i white balls. At each step, we draw one ball from each urn and place the ball drawn from the first urn into the second, and conversely with the ball from the second urn. Let X_n denote the state of the system after the nth step. Explain why $\{X_n, n = 0, 1, 2, \ldots\}$ is a Markov chain and calculate its transition probability matrix.

2. Each individual in a population independently has a random number of offspring that is Poisson distributed with mean λ. Those initially present constitute the zeroth generation. Offspring of zeroth generation people constitute the first generation; their offspring constitute the second generation, and so on. If X_n denotes the size of generation n, is $\{X_n, n \geq 0\}$ a Markov chain. If it is, give its transition probabilities $P_{i,j}$; if it is not, explain why it is not.

3. There are k players, with player i having value $v_i > 0$, $i = 1, \ldots, k$. In every period, two of the players play a game, while the other $k - 2$ wait in an ordered line. The loser of a game joins the end of the line, and the winner then plays a new game against the player who is first in line. Whenever i and j play, i wins with probability $\frac{v_i}{v_i+v_j}$.

(a) Define a Markov chain that is useful in analyzing this model.
(b) How many states does the Markov chain have?
(c) Give the transition probabilities of the chain.

4. Let **P** and **Q** be transition probability matrices on states $1, \ldots, m$, with respective transition probabilities $P_{i,j}$ and $Q_{i,j}$. Consider processes $\{X_n, n \geq 0\}$ and $\{Y_n, n \geq 0\}$ defined as follows:

(a) $X_0 = 1$. A coin that comes up heads with probability p is then flipped. If the coin lands heads, the subsequent states $X_1, X_2, \ldots$, are obtained by using the transition probability matrix **P**; if it lands tails, the subsequent states $X_1, X_2, \ldots$, are obtained by using the transition probability matrix **Q**. (In other words, if the coin lands heads (tails) then the sequence of states is a Markov chain with transition probability matrix **P** (**Q**).) Is $\{X_n, n \geq 0\}$ a Markov chain. If it is, give its transition probabilities. If it is not, tell why not.

(b) $Y_0 = 1$. If the current state is i, then the next state is determined by first flipping a coin that comes up heads with probability p. If the coin lands heads then the next state is j with probability $P_{i,j}$; if it lands tails then the next state is j with probability $Q_{i,j}$. Is $\{Y_n, n \geq 0\}$ a Markov chain. If it is, give its transition probabilities. If it is not, tell why not.

5. A Markov chain $\{X_n, n \geqslant 0\}$ with states 0, 1, 2, has the transition probability matrix

$$\begin{bmatrix} \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ 0 & \frac{1}{3} & \frac{2}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}$$

If $P\{X_0 = 0\} = P\{X_0 = 1\} = \frac{1}{4}$, find $E[X_3]$.

6. Let the transition probability matrix of a two-state Markov chain be given, as in Example 4.2, by

$$\mathbf{P} = \begin{Vmatrix} p & 1-p \\ 1-p & p \end{Vmatrix}$$

Show by mathematical induction that

$$\mathbf{P}^{(n)} = \begin{Vmatrix} \frac{1}{2} + \frac{1}{2}(2p-1)^n & \frac{1}{2} - \frac{1}{2}(2p-1)^n \\ \frac{1}{2} - \frac{1}{2}(2p-1)^n & \frac{1}{2} + \frac{1}{2}(2p-1)^n \end{Vmatrix}$$

7. In Example 4.4 suppose that it has rained neither yesterday nor the day before yesterday. What is the probability that it will rain tomorrow?

8. An urn initially contains 2 balls, one of which is red and the other blue. At each stage a ball is randomly selected. If the selected ball is red, then it is replaced with a red ball with probability .7 or with a blue ball with probability .3; if the selected ball is blue, then it is equally likely to be replaced by either a red or blue ball.

(a) Let X_n equal 1 if the nth ball selected is red, and let it equal 0 otherwise. Is $\{X_n, n \geq 1\}$ a Markov chain? If so, give its transition probability matrix.

(b) Let Y_n denote the number of red balls in the urn immediately before the nth ball is selected. Is $\{Y_n, n \geq 1\}$ a Markov chain? If so, give its transition probability matrix.

(c) Find the probability that the second ball selected is red.

(d) Find the probability that the fourth ball selected is red.

***9.** In a sequence of independent flips of a coin that comes up heads with probability .6, what is the probability that there is a run of three consecutive heads within the first 10 flips?

10. In Example 4.3, Gary is currently in a cheerful mood. What is the probability that he is not in a glum mood on any of the following three days?

11. In Example 4.13, give the transition probabilities of the Y_n Markov chain in terms of the transition probabilities $P_{i,j}$ of the X_n chain.

12. For a Markov chain $\{X_n, n \geqslant 0\}$ with transition probabilities $P_{i,j}$, consider the conditional probability that $X_n = m$ given that the chain started at time 0 in state i and has not yet entered state r by time n, where r is a specified state not equal to either i or m. We are interested in whether this conditional probability is equal to the n stage transition probability of a Markov chain whose state space does not include state r and whose transition probabilities are

$$Q_{i,j} = \frac{P_{i,j}}{1 - P_{i,r}}, \quad i, j \neq r$$

Either prove the equality

$$P\{X_n = m | X_0 = i, X_k \neq r, k = 1, \ldots, n\} = Q^n_{i,m}$$

or construct a counterexample.

13. Let $\mathbf{P}$ be the transition probability matrix of a Markov chain. Argue that if for some positive integer r, $\mathbf{P}^r$ has all positive entries, then so does $\mathbf{P}^n$, for all integers $n \geqslant r$.

14. Specify the classes of the following Markov chains, and determine whether they are transient or recurrent:

$$\mathbf{P}_1 = \begin{Vmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{Vmatrix}, \qquad \mathbf{P}_2 = \begin{Vmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \end{Vmatrix},$$

$$\mathbf{P}_3 = \begin{Vmatrix} \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{Vmatrix}, \qquad \mathbf{P}_4 = \begin{Vmatrix} \frac{1}{4} & \frac{3}{4} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{3} & \frac{2}{3} & 0 \\ 1 & 0 & 0 & 0 & 0 \end{Vmatrix}$$

15. Prove that if the number of states in a Markov chain is M, and if state j can be reached from state i, then it can be reached in M steps or less.

***16.** Show that if state i is recurrent and state i does not communicate with state j, then $P_{ij} = 0$. This implies that once a process enters a recurrent class of states it can never leave that class. For this reason, a recurrent class is often referred to as a *closed* class.

17. For the random walk of Example 4.19 use the strong law of large numbers to give another proof that the Markov chain is transient when $p \neq \frac{1}{2}$.

Hint: Note that the state at time n can be written as $\sum_{i=1}^{n} Y_i$ where the Y_is are independent and $P\{Y_i = 1\} = p = 1 - P\{Y_i = -1\}$. Argue that if $p > \frac{1}{2}$, then, by the strong law of large numbers, $\sum_1^n Y_i \to \infty$ as $n \to \infty$ and hence the initial state 0 can be visited only finitely often, and hence must be transient. A similar argument holds when $p < \frac{1}{2}$.

18. Coin 1 comes up heads with probability 0.6 and coin 2 with probability 0.5. A coin is continually flipped until it comes up tails, at which time that coin is put aside and we start flipping the other one.

(a) What proportion of flips use coin 1?

(b) If we start the process with coin 1 what is the probability that coin 2 is used on the fifth flip?

(c) What proportion of flips land heads?

19. For Example 4.4, calculate the proportion of days that it rains.

20. A transition probability matrix $\mathbf{P}$ is said to be doubly stochastic if the sum over each column equals one; that is,

$$\sum_i P_{ij} = 1, \quad \text{for all } j$$

If such a chain is irreducible and consists of $M + 1$ states $0, 1, \ldots, M$, show that the long-run proportions are given by

$$\pi_j = \frac{1}{M+1}, \quad j = 0, 1, \ldots, M$$

***21.** A DNA nucleotide has any of four values. A standard model for a mutational change of the nucleotide at a specific location is a Markov chain model that supposes that in going from period to period the nucleotide does not change with probability $1 - 3\alpha$, and if it does change then it is equally likely to change to any of the other three values, for some $0 < \alpha < \frac{1}{3}$.

(a) Show that $P_{1,1}^n = \frac{1}{4} + \frac{3}{4}(1-4\alpha)^n$.
(b) What is the long-run proportion of time the chain is in each state?

22. Let Y_n be the sum of n independent rolls of a fair die. Find

$$\lim_{n\to\infty} P\{Y_n \text{ is a multiple of 13}\}$$

Hint: Define an appropriate Markov chain and apply the results of Exercise 20.

23. In a good weather year the number of storms is Poisson distributed with mean 1; in a bad year it is Poisson distributed with mean 3. Suppose that any year's weather conditions depends on past years only through the previous year's condition. Suppose that a good year is equally likely to be followed by either a good or a bad year, and that a bad year is twice as likely to be followed by a bad year as by a good year. Suppose that last year—call it year 0—was a good year.
(a) Find the expected total number of storms in the next two years (that is, in years 1 and 2).
(b) Find the probability there are no storms in year 3.
(c) Find the long-run average number of storms per year.
(d) Find the proportion of years that have no storms.

24. Consider three urns, one colored red, one white, and one blue. The red urn contains 1 red and 4 blue balls; the white urn contains 3 white balls, 2 red balls, and 2 blue balls; the blue urn contains 4 white balls, 3 red balls, and 2 blue balls. At the initial stage, a ball is randomly selected from the red urn and then returned to that urn. At every subsequent stage, a ball is randomly selected from the urn whose color is the same as that of the ball previously selected and is then returned to that urn. In the long run, what proportion of the selected balls are red? What proportion are white? What proportion are blue?

25. Each morning an individual leaves his house and goes for a run. He is equally likely to leave either from his front or back door. Upon leaving the house, he chooses a pair of running shoes (or goes running barefoot if there are no shoes at the door from which he departed). On his return he is equally likely to enter, and leave his running shoes, either by the front or back door. If he owns a total of k pairs of running shoes, what proportion of the time does he run barefooted?

26. Consider the following approach to shuffling a deck of n cards. Starting with any initial ordering of the cards, one of the numbers $1, 2, \ldots, n$ is randomly chosen in such a manner that each one is equally likely to be selected. If number i is chosen, then we take the card that is in position i and put it on top of the deck—that is, we put that card in position 1. We then repeatedly perform the same operation. Show that, in the limit, the deck is perfectly shuffled in the sense that the resultant ordering is equally likely to be any of the $n!$ possible orderings.

***27.** Each individual in a population of size N is, in each period, either active or inactive. If an individual is active in a period then, independent of all else, that individual will be active in the next period with probability α. Similarly, if an

individual is inactive in a period then, independent of all else, that individual will be inactive in the next period with probability β. Let X_n denote the number of individuals that are active in period n.

(a) Argue that $X_n, n \geqslant 0$ is a Markov chain.
(b) Find $E[X_n|X_0 = i]$.
(c) Derive an expression for its transition probabilities.
(d) Find the long-run proportion of time that exactly j people are active.

Hint for (d): Consider first the case where $N = 1$.

28. Every time that the team wins a game, it wins its next game with probability 0.8; every time it loses a game, it wins its next game with probability 0.3. If the team wins a game, then it has dinner together with probability 0.7, whereas if the team loses then it has dinner together with probability 0.2. What proportion of games result in a team dinner?

29. An organization has N employees where N is a large number. Each employee has one of three possible job classifications and changes classifications (independently) according to a Markov chain with transition probabilities

$$\begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.2 & 0.6 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{bmatrix}$$

What percentage of employees are in each classification?

30. Three out of every four trucks on the road are followed by a car, while only one out of every five cars is followed by a truck. What fraction of vehicles on the road are trucks?

31. A certain town never has two sunny days in a row. Each day is classified as being either sunny, cloudy (but dry), or rainy. If it is sunny one day, then it is equally likely to be either cloudy or rainy the next day. If it is rainy or cloudy one day, then there is one chance in two that it will be the same the next day, and if it changes then it is equally likely to be either of the other two possibilities. In the long run, what proportion of days are sunny? What proportion are cloudy?

***32.** Each of two switches is either on or off during a day. On day n, each switch will independently be on with probability

$$[1 + \text{number of on switches during day } n - 1]/4$$

For instance, if both switches are on during day $n - 1$, then each will independently be on during day n with probability $3/4$. What fraction of days are both switches on? What fraction are both off?

33. Two players are playing a sequence of points, which begin when one of the players serves. Suppose that player 1 wins each point she serves with probability p, and wins each point her opponent serves with probability q. Suppose the winner of a point becomes the server of the next point.

(a) Find the proportion of points that are won by player 1.
(b) Find the proportion of time that player 1 is the server.

34. A flea moves around the vertices of a triangle in the following manner: Whenever it is at vertex i it moves to its clockwise neighbor vertex with probability p_i and to the counterclockwise neighbor with probability $q_i = 1 - p_i, i = 1, 2, 3$.

(a) Find the proportion of time that the flea is at each of the vertices.

(b) How often does the flea make a counterclockwise move that is then followed by five consecutive clockwise moves?

35. Consider a Markov chain with states 0, 1, 2, 3, 4. Suppose $P_{0,4} = 1$; and suppose that when the chain is in state $i, i > 0$, the next state is equally likely to be any of the states $0, 1, \ldots, i - 1$. Find the limiting probabilities of this Markov chain.

36. The state of a process changes daily according to a two-state Markov chain. If the process is in state i during one day, then it is in state j the following day with probability $P_{i,j}$, where

$$P_{0,0} = 0.4, \quad P_{0,1} = 0.6, \quad P_{1,0} = 0.2, \quad P_{1,1} = 0.8$$

Every day a message is sent. If the state of the Markov chain that day is i then the message sent is "good" with probability p_i and is "bad" with probability $q_i = 1 - p_i, i = 0, 1$

(a) If the process is in state 0 on Monday, what is the probability that a good message is sent on Tuesday?

(b) If the process is in state 0 on Monday, what is the probability that a good message is sent on Friday?

(c) In the long run, what proportion of messages are good?

(d) Let Y_n equal 1 if a good message is sent on day n and let it equal 2 otherwise. Is $\{Y_n, n \geqslant 1\}$ a Markov chain? If so, give its transition probability matrix. If not, briefly explain why not.

37. Show that the stationary probabilities for the Markov chain having transition probabilities $P_{i,j}$ are also the stationary probabilities for the Markov chain whose transition probabilities $Q_{i,j}$ are given by

$$Q_{i,j} = P_{i,j}^k$$

for any specified positive integer k.

38. Capa plays either one or two chess games every day, with the number of games that she plays on successive days being a Markov chain with transition probabilities

$$P_{1,1} = .2, \quad P_{1,2} = .8 \qquad P_{2,1} = .4, \quad P_{2,2} = .6$$

Capa wins each game with probability p. Suppose she plays two games on Monday.

(a) What is the probability that she wins all the games she plays on Tuesday?

(b) What is the expected number of games that she plays on Wednesday?

(c) In the long run, on what proportion of days does Capa win all her games.

39. Consider the one-dimensional symmetric random walk of Example 4.19, which was shown in that example to be recurrent. Let π_i denote the long-run proportion of time that the chain is in state i.
(a) Argue that $\pi_i = \pi_0$ for all i.
(b) Show that $\sum_i \pi_i \neq 1$.
(c) Conclude that this Markov chain is null recurrent, and thus all $\pi_i = 0$.

40. A particle moves on 12 points situated on a circle. At each step it is equally likely to move one step in the clockwise or in the counterclockwise direction. Find the mean number of steps for the particle to return to its starting position.

***41.** Consider a Markov chain with states equal to the nonnegative integers, and suppose its transition probabilities satisfy $P_{i,j} = 0$ if $j \leq i$. Assume $X_0 = 0$, and let e_j be the probability that the Markov chain is ever in state j. (Note that $e_0 = 1$ because $X_0 = 0$.) Argue that for $j > 0$

$$e_j = \sum_{i=0}^{j-1} e_i P_{i,j}$$

If $P_{i,i+k} = 1/3, k = 1, 2, 3$, find e_i for $i = 1, \ldots, 10$.

42. Let A be a set of states, and let A^c be the remaining states.
(a) What is the interpretation of

$$\sum_{i \in A} \sum_{j \in A^c} \pi_i P_{ij}?$$

(b) What is the interpretation of

$$\sum_{i \in A^c} \sum_{j \in A} \pi_i P_{ij}?$$

(c) Explain the identity

$$\sum_{i \in A} \sum_{j \in A^c} \pi_i P_{ij} = \sum_{i \in A^c} \sum_{j \in A} \pi_i P_{ij}$$

43. Each day, one of n possible elements is requested, the ith one with probability $P_i, i \geqslant 1, \sum_1^n P_i = 1$. These elements are at all times arranged in an ordered list that is revised as follows: The element selected is moved to the front of the list with the relative positions of all the other elements remaining unchanged. Define the state at any time to be the list ordering at that time and note that there are $n!$ possible states.
(a) Argue that the preceding is a Markov chain.
(b) For any state $i_1, \ldots, i_n$ (which is a permutation of $1, 2, \ldots, n$), let $\pi(i_1, \ldots, i_n)$ denote the limiting probability. In order for the state to be $i_1, \ldots, i_n$, it is necessary for the last request to be for i_1, the last non-i_1 request for i_2, the last non-i_1 or i_2 request for i_3, and so on. Hence, it

appears intuitive that

$$\pi(i_1, \ldots, i_n) = P_{i_1} \frac{P_{i_2}}{1 - P_{i_1}} \frac{P_{i_3}}{1 - P_{i_1} - P_{i_2}} \cdots \frac{P_{i_{n-1}}}{1 - P_{i_1} - \cdots - P_{i_{n-2}}}$$

Verify when $n = 3$ that the preceding are indeed the limiting probabilities.

44. Suppose that a population consists of a fixed number, say, m, of genes in any generation. Each gene is one of two possible genetic types. If exactly i (of the m) genes of any generation are of type 1, then the next generation will have j type 1 (and $m - j$ type 2) genes with probability

$$\binom{m}{j} \left(\frac{i}{m}\right)^j \left(\frac{m-i}{m}\right)^{m-j}, \quad j = 0, 1, \ldots, m$$

Let X_n denote the number of type 1 genes in the nth generation, and assume that $X_0 = i$.

(a) Find $E[X_n]$.

(b) What is the probability that eventually all the genes will be type 1?

45. Consider an irreducible finite Markov chain with states $0, 1, \ldots, N$.

(a) Starting in state i, what is the probability the process will ever visit state j? Explain!

(b) Let $x_i = P\{\text{visit state } N \text{ before state } 0 | \text{start in } i\}$. Compute a set of linear equations that the x_i satisfy, $i = 0, 1, \ldots, N$.

(c) If $\sum_j j P_{ij} = i$ for $i = 1, \ldots, N - 1$, show that $x_i = i/N$ is a solution to the equations in part (b).

46. An individual possesses r umbrellas that he employs in going from his home to office, and vice versa. If he is at home (the office) at the beginning (end) of a day and it is raining, then he will take an umbrella with him to the office (home), provided there is one to be taken. If it is not raining, then he never takes an umbrella. Assume that, independent of the past, it rains at the beginning (end) of a day with probability p.

(a) Define a Markov chain with $r + 1$ states, which will help us to determine the proportion of time that our man gets wet. (*Note*: He gets wet if it is raining, and all umbrellas are at his other location.)

(b) Show that the limiting probabilities are given by

$$\pi_i = \begin{cases} \dfrac{q}{r+q}, & \text{if } i = 0 \\ \dfrac{1}{r+q}, & \text{if } i = 1, \ldots, r \end{cases} \qquad \text{where } q = 1 - p$$

(c) What fraction of time does our man get wet?

(d) When $r = 3$, what value of p maximizes the fraction of time he gets wet

***47.** Let $\{X_n, n \geqslant 0\}$ denote an ergodic Markov chain with limiting probabilities π_i. Define the process $\{Y_n, n \geqslant 1\}$ by $Y_n = (X_{n-1}, X_n)$. That is, Y_n keeps track of

the last two states of the original chain. Is $\{Y_n, n \geqslant 1\}$ a Markov chain? If so, determine its transition probabilities and find

$$\lim_{n\to\infty} P\{Y_n = (i, j)\}$$

48. Consider a Markov chain in steady state. Say that a k length run of zeroes ends at time m if

$$X_{m-k-1} \neq 0, \quad X_{m-k} = X_{m-k+1} = \ldots = X_{m-1} = 0, X_m \neq 0$$

Show that the probability of this event is $\pi_0(P_{0,0})^{k-1}(1 - P_{0,0})^2$, where π_0 is the limiting probability of state 0.

49. Consider a Markov chain with states 1, 2, 3 having transition probability matrix

$$\begin{pmatrix} .5 & .3 & .2 \\ 0 & .4 & .6 \\ .8 & 0 & .2 \end{pmatrix}$$

(a) If the chain is currently in state 1, find the probability that after two transitions it will be in state 2.

(b) Suppose you receive a reward $r(i) = i^2$ whenever the Markov chain is in state i, $i = 1, 2, 3$. Find your long run average reward per unit time.

Let N_i denote the number of transitions, starting in state i, until the Markov chain enters state 3.

(c) Find $E[N_1]$.

(d) Find $P(N_1 \leq 4)$.

(e) Find $P(N_1 = 4)$.

50. A Markov chain with states $1, \ldots, 6$ has transition probability matrix

$$\begin{pmatrix} .2 & .4 & 0 & .3 & 0 & .1 \\ .1 & .3 & 0 & .4 & 0 & .2 \\ 0 & 0 & .3 & .7 & 0 & 0 \\ 0 & 0 & .6 & .4 & 0 & 0 \\ 0 & 0 & 0 & 0 & .5 & .5 \\ 0 & 0 & 0 & 0 & .2 & .8 \end{pmatrix}$$

(a) Give the classes and tell which are recurrent and which are transient.

(b) Find $\lim_{n\to\infty} P^n_{1,2}$.

(c) Find $\lim_{n\to\infty} P^n_{5,6}$.

(d) Find $\lim_{n\to\infty} P^n_{1,3}$.

51. In Example 4.3, Gary is in a cheerful mood today. Find the expected number of days until he has been glum for three consecutive days.

52. A taxi driver provides service in two zones of a city. Fares picked up in zone A will have destinations in zone A with probability 0.6 or in zone B with probability 0.4. Fares picked up in zone B will have destinations in zone A with probability 0.3 or in zone B with probability 0.7. The driver's expected

profit for a trip entirely in zone A is 6; for a trip entirely in zone B is 8; and for a trip that involves both zones is 12. Find the taxi driver's average profit per trip.

53. Find the average premium received per policyholder of the insurance company of Example 4.29 if $\lambda = 1/4$ for one-third of its clients, and $\lambda = 1/2$ for two-thirds of its clients.

54. Consider the Ehrenfest urn model in which M molecules are distributed between two urns, and at each time point one of the molecules is chosen at random and is then removed from its urn and placed in the other one. Let X_n denote the number of molecules in urn 1 after the nth switch and let $\mu_n = E[X_n]$. Show that

(a) $\mu_{n+1} = 1 + (1 - 2/M)\mu_n$.

(b) Use (a) to prove that

$$\mu_n = \frac{M}{2} + \left(\frac{M-2}{M}\right)^n \left(E[X_0] - \frac{M}{2}\right)$$

55. Consider a population of individuals each of whom possesses two genes that can be either type A or type a. Suppose that in outward appearance type A is dominant and type a is recessive. (That is, an individual will have only the outward characteristics of the recessive gene if its pair is aa.) Suppose that the population has stabilized, and the percentages of individuals having respective gene pairs AA, aa, and Aa are p, q, and r. Call an individual dominant or recessive depending on the outward characteristics it exhibits. Let S_{11} denote the probability that an offspring of two dominant parents will be recessive; and let S_{10} denote the probability that the offspring of one dominant and one recessive parent will be recessive. Compute S_{11} and S_{10} to show that $S_{11} = S_{10}^2$. (The quantities S_{10} and S_{11} are known in the genetics literature as *Snyder's ratios*.)

56. Suppose that on each play of the game a gambler either wins 1 with probability p or loses 1 with probability $1 - p$. The gambler continues betting until she or he is either up n or down m. What is the probability that the gambler quits a winner?

57. A particle moves among $n + 1$ vertices that are situated on a circle in the following manner. At each step it moves one step either in the clockwise direction with probability p or the counterclockwise direction with probability $q = 1 - p$. Starting at a specified state, call it state 0, let T be the time of the first return to state 0. Find the probability that all states have been visited by time T.

Hint: Condition on the initial transition and then use results from the gambler's ruin problem.

58. In the gambler's ruin problem of Section 4.5.1, suppose the gambler's fortune is presently i, and suppose that we know that the gambler's fortune will eventually reach N (before it goes to 0). Given this information, show that the

probability he wins the next gamble is

$$\begin{cases} \dfrac{p[1-(q/p)^{i+1}]}{1-(q/p)^i}, & \text{if } p \neq \frac{1}{2} \\ \dfrac{i+1}{2i}, & \text{if } p = \frac{1}{2} \end{cases}$$

Hint: The probability we want is

$$P\{X_{n+1} = i+1 | X_n = i, \lim_{m\to\infty} X_m = N\} = \frac{P\{X_{n+1} = i+1, \lim_m X_m = N | X_n = i\}}{P\{\lim_m X_m = N | X_n = i\}}$$

59. For the gambler's ruin model of Section 4.5.1, let M_i denote the mean number of games that must be played until the gambler either goes broke or reaches a fortune of N, given that he starts with i, $i = 0, 1, \ldots, N$. Show that M_i satisfies

$$M_0 = M_N = 0; \quad M_i = 1 + pM_{i+1} + qM_{i-1}, \quad i = 1, \ldots, N-1$$

Solve these equations to obtain

$$\begin{aligned} M_i &= i(N-i), && \text{if } p = \tfrac{1}{2} \\ &= \frac{i}{q-p} - \frac{N}{q-p}\frac{1-(q/p)^i}{1-(q/p)^N}, && \text{if } p \neq \tfrac{1}{2} \end{aligned}$$

60. The following is the transition probability matrix of a Markov chain with states 1, 2, 3, 4

$$\mathbf{P} = \begin{pmatrix} .4 & .3 & .2 & .1 \\ .2 & .2 & .2 & .4 \\ .25 & .25 & .5 & 0 \\ .2 & .1 & .4 & .3 \end{pmatrix}$$

If $X_0 = 1$

(a) find the probability that state 3 is entered before state 4;

(b) find the mean number of transitions until either state 3 or state 4 is entered.

61. Suppose in the gambler's ruin problem that the probability of winning a bet depends on the gambler's present fortune. Specifically, suppose that α_i is the probability that the gambler wins a bet when his or her fortune is i. Given that the gambler's initial fortune is i, let $P(i)$ denote the probability that the gambler's fortune reaches N before 0.

(a) Derive a formula that relates $P(i)$ to $P(i-1)$ and $P(i+1)$.

(b) Using the same approach as in the gambler's ruin problem, solve the equation of part (a) for $P(i)$.

(c) Suppose that i balls are initially in urn 1 and $N - i$ are in urn 2, and suppose that at each stage one of the N balls is randomly chosen, taken from whichever urn it is in, and placed in the other urn. Find the probability that the first urn becomes empty before the second.

***62.** Consider the particle from Exercise 57. What is the expected number of steps the particle takes to return to the starting position? What is the probability that all other positions are visited before the particle returns to its starting state?

63. For the Markov chain with states 1, 2, 3, 4 whose transition probability matrix $\mathbf{P}$ is as specified below find f_{i3} and s_{i3} for $i = 1, 2, 3$.

$$\mathbf{P} = \begin{bmatrix} 0.4 & 0.2 & 0.1 & 0.3 \\ 0.1 & 0.5 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.2 & 0.1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

64. Consider a branching process having $\mu < 1$. Show that if $X_0 = 1$, then the expected number of individuals that ever exist in this population is given by $1/(1 - \mu)$. What if $X_0 = n$?

65. In a branching process having $X_0 = 1$ and $\mu > 1$, prove that π_0 is the *smallest* positive number satisfying Eq. (4.20).

Hint: Let π be any solution of $\pi = \sum_{j=0}^{\infty} \pi^j P_j$. Show by mathematical induction that $\pi \geqslant P\{X_n = 0\}$ for all n, and let $n \to \infty$. In using the induction argue that

$$P\{X_n = 0\} = \sum_{j=0}^{\infty} (P\{X_{n-1} = 0\})^j P_j$$

66. For a branching process, calculate π_0 when

(a) $P_0 = \frac{1}{4}, P_2 = \frac{3}{4}$.

(b) $P_0 = \frac{1}{4}, P_1 = \frac{1}{2}, P_2 = \frac{1}{4}$.

(c) $P_0 = \frac{1}{6}, P_1 = \frac{1}{2}, P_3 = \frac{1}{3}$.

67. At all times, an urn contains N balls—some white balls and some black balls. At each stage, a coin having probability $p, 0 < p < 1$, of landing heads is flipped. If heads appears, then a ball is chosen at random from the urn and is replaced by a white ball; if tails appears, then a ball is chosen from the urn and is replaced by a black ball. Let X_n denote the number of white balls in the urn after the nth stage.

(a) Is $\{X_n, n \geqslant 0\}$ a Markov chain? If so, explain why.

(b) What are its classes? What are their periods? Are they transient or recurrent?

(c) Compute the transition probabilities P_{ij}.

(d) Let $N = 2$. Find the proportion of time in each state.

(e) Based on your answer in part (d) and your intuition, guess the answer for the limiting probability in the general case.

(f) Prove your guess in part (e) either by showing that Theorem (4.1) is satisfied or by using the results of Example 4.37.

(g) If $p = 1$, what is the expected time until there are only white balls in the urn if initially there are i white and $N - i$ black?

***68.** **(a)** Show that the limiting probabilities of the reversed Markov chain are the same as for the forward chain by showing that they satisfy the equations

$$\pi_j = \sum_i \pi_i Q_{ij}$$

(b) Give an intuitive explanation for the result of part (a).

69. M balls are initially distributed among m urns. At each stage one of the balls is selected at random, taken from whichever urn it is in, and then placed, at random, in one of the other $m - 1$ urns. Consider the Markov chain whose state at any time is the vector $(n_1, \ldots, n_m)$ where n_i denotes the number of balls in urn i. Guess at the limiting probabilities for this Markov chain and then verify your guess and show at the same time that the Markov chain is time reversible.

70. A total of m white and m black balls are distributed among two urns, with each urn containing m balls. At each stage, a ball is randomly selected from each urn and the two selected balls are interchanged. Let X_n denote the number of black balls in urn 1 after the nth interchange.

(a) Give the transition probabilities of the Markov chain $X_n, n \geqslant 0$.

(b) Without any computations, what do you think are the limiting probabilities of this chain?

(c) Find the limiting probabilities and show that the stationary chain is time reversible.

71. It follows from Theorem 4.2 that for a time reversible Markov chain

$$P_{ij} P_{jk} P_{ki} = P_{ik} P_{kj} P_{ji}, \quad \text{for all } i, j, k$$

It turns out that if the state space is finite and $P_{ij} > 0$ for all i, j, then the preceding is also a sufficient condition for time reversibility. (That is, in this case, we need only check Eq. (4.26) for paths from i to i that have only two intermediate states.) Prove this.

Hint: Fix i and show that the equations

$$\pi_j P_{jk} = \pi_k P_{kj}$$

are satisfied by $\pi_j = c P_{ij} / P_{ji}$, where c is chosen so that $\sum_j \pi_j = 1$.

72. For a time reversible Markov chain, argue that the rate at which transitions from i to j to k occur must equal the rate at which transitions from k to j to i occur.

73. There are k players, with player i having value $v_i > 0$, $i = 1, \ldots, k$. In every period, two of the players play a game. Whoever wins then plays the next game against a randomly chosen one of the other $k - 1$ players (including the one

who has just lost). Suppose that whenever i and j play, i wins with probability $\frac{v_i}{v_i+v_j}$. Let X_n denote the winner of game n.

(a) Give the transition probabilities of the Markov chain $\{X_n, n \geq 1\}$.
(b) Give the stationarity equations that are uniquely satisfied by the π_j.
(c) Give the time reversibility equations.
(d) Find the proportion of all games that are won by j, $j = 1, \ldots, k$.
(e) Find the proportion of games that involve player j as one of the contestants.

74. A group of n processors is arranged in an ordered list. When a job arrives, the first processor in line attempts it; if it is unsuccessful, then the next in line tries it; if it too is unsuccessful, then the next in line tries it, and so on. When the job is successfully processed or after all processors have been unsuccessful, the job leaves the system. At this point we are allowed to reorder the processors, and a new job appears. Suppose that we use the one-closer reordering rule, which moves the processor that was successful one closer to the front of the line by interchanging its position with the one in front of it. If all processors were unsuccessful (or if the processor in the first position was successful), then the ordering remains the same. Suppose that each time processor i attempts a job then, independently of anything else, it is successful with probability p_i.

(a) Define an appropriate Markov chain to analyze this model.
(b) Show that this Markov chain is time reversible.
(c) Find the long-run probabilities.

75. A Markov chain is said to be a tree process if

(i) $P_{ij} > 0$ whenever $P_{ji} > 0$,
(ii) for every pair of states i and j, $i \neq j$, there is a unique sequence of distinct states $i = i_0, i_1, \ldots, i_{n-1}, i_n = j$ such that

$$P_{i_k, i_{k+1}} > 0, \quad k = 0, 1, \ldots, n-1$$

In other words, a Markov chain is a tree process if for every pair of distinct states i and j there is a unique way for the process to go from i to j without reentering a state (and this path is the reverse of the unique path from j to i). Argue that an ergodic tree process is time reversible.

76. On a chessboard compute the expected number of plays it takes a knight, starting in one of the four corners of the chessboard, to return to its initial position if we assume that at each play it is equally likely to choose any of its legal moves. (No other pieces are on the board.)

Hint: Make use of Example 4.38.

77. In a Markov decision problem, another criterion often used, different than the expected average return per unit time, is that of the expected discounted return. In this criterion we choose a number α, $0 < \alpha < 1$, and try to choose a policy so as to maximize $E[\sum_{i=0}^{\infty} \alpha^i R(X_i, a_i)]$ (that is, rewards at time n are discounted at rate α^n). Suppose that the initial state is chosen according to the probabilities b_i. That is,

$$P\{X_0 = i\} = b_i, \quad i = 1, \ldots, n$$

For a given policy $\boldsymbol{\beta}$ let y_{ja} denote the expected discounted time that the process is in state j and action a is chosen. That is,

$$y_{ja} = E_{\boldsymbol{\beta}}\left[\sum_{n=0}^{\infty} \alpha^n I_{\{X_n=j, a_n=a\}}\right]$$

where for any event A the indicator variable I_A is defined by

$$I_A = \begin{cases} 1, & \text{if } A \text{ occurs} \\ 0, & \text{otherwise} \end{cases}$$

(a) Show that

$$\sum_a y_{ja} = E\left[\sum_{n=0}^{\infty} \alpha^n I_{\{X_n=j\}}\right]$$

or, in other words, $\sum_a y_{ja}$ is the expected discounted time in state j under $\boldsymbol{\beta}$.

(b) Show that

$$\sum_j \sum_a y_{ja} = \frac{1}{1-\alpha},$$

$$\sum_a y_{ja} = b_j + \alpha \sum_i \sum_a y_{ia} P_{ij}(a)$$

Hint: For the second equation, use the identity

$$I_{\{X_{n+1}=j\}} = \sum_i \sum_a I_{\{X_n=i, a_n=a\}} I_{\{X_{n+1}=j\}}$$

Take expectations of the preceding to obtain

$$E\left[I_{X_{n+1}=j\}}\right] = \sum_i \sum_a E\left[I_{\{X_n=i, a_n=a\}}\right] P_{ij}(a)$$

(c) Let $\{y_{ja}\}$ be a set of numbers satisfying

$$\begin{aligned} &\sum_j \sum_a y_{ja} = \frac{1}{1-\alpha}, \\ &\sum_a y_{ja} = b_j + \alpha \sum_i \sum_a y_{ia} P_{ij}(a) \end{aligned} \tag{4.38}$$

Argue that y_{ja} can be interpreted as the expected discounted time that the process is in state j and action a is chosen when the initial state is chosen

according to the probabilities b_j and the policy $\boldsymbol{\beta}$, given by

$$\beta_i(a) = \frac{y_{ia}}{\sum_a y_{ia}}$$

is employed.

Hint: Derive a set of equations for the expected discounted times when policy $\boldsymbol{\beta}$ is used and show that they are equivalent to Eq. (4.38).

(d) Argue that an optimal policy with respect to the expected discounted return criterion can be obtained by first solving the linear program

$$\begin{aligned} \text{maximize} \quad & \sum_j \sum_a y_{ja} R(j,a), \\ \text{such that} \quad & \sum_j \sum_a y_{ja} = \frac{1}{1-\alpha}, \\ & \sum_a y_{ja} = b_j + \alpha \sum_i \sum_a y_{ia} P_{ij}(a), \\ & y_{ja} \geqslant 0, \quad \text{all } j, a; \end{aligned}$$

and then defining the policy β^* by

$$\beta_i^*(a) = \frac{y_{ia}^*}{\sum_a y_{ia}^*}$$

where the y_{ja}^* are the solutions of the linear program.

78. For the Markov chain of Exercise 5, suppose that $p(s|j)$ is the probability that signal s is emitted when the underlying Markov chain state is j, $j = 0, 1, 2$.

(a) What proportion of emissions are signal s?

(b) What proportion of those times in which signal s is emitted is 0 the underlying state?

79. In Example 4.45, what is the probability that the first 4 items produced are all acceptable?

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